

Emotion-Aware Recommender Systems: A Comprehensive Review, Challenges, and Future Directions

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Emotion-aware recommender systems have received a lot of interest in recent years because of their ability to improve the user experience by adapting recommendations to users' emotional states. This review paper conducts a thorough examination of emotion-aware recommender systems, categorizing them according to the datasets used (video, text, audio, image and physiological), the application domains (movie, music, social media and news), the emotion categorization and the methodology used. We investigate multiple issues facing these systems, such as accurately detecting and interpreting emotions, integrating multimodal data, and ethical concerns about user privacy and emotional manipulation. In addition, we explore potential future directions for emotion-aware recommender systems.

CCS Concepts: • **Information systems** → **Personalization; Recommender systems; Multimedia and multimodal retrieval.**

Additional Key Words and Phrases: Recommender Systems, Machine Learning, Deep Learning, Emotion Recognition, Sentiment Analysis, Personalization, Collaborative Filtering, Content-based Filtering, Multimodal Learning, User Modeling, Review Mining, Natural Language Processing

1 Introduction

In this modern era, people are inundated with a huge amount of choices and products, a sharp contrast to the past, when selections were far more limited. This explosion of available options has transformed the way people make purchasing decisions and discover new interests. Whether it is selecting a book to read, purchasing a product online, receiving personalized medication recommendations, or finding the best places to eat a delicious meal, the sheer volume of choices can be overwhelming [161].

Recommender systems have been developed as a crucial component of our digital lives, significantly shaping how we discover and engage with content on various platforms. These systems have transformed e-Commerce, guiding consumers to products that they are likely to enjoy and purchase [3]. In the field of entertainment, they help users find movies, music, and shows that align with their tastes. Social networks leverage recommender systems to improve the user experience by suggesting friends, groups, and content. Location-based recommendations help travelers discover tourist attractions and dining options. News platforms use these systems to present readers with articles that match their interests while filtering out content that may be less relevant [106].

The impact of recommender systems extends beyond convenience; they play a critical role in filtering information, ensuring users receive content tailored to their or the company's preferences and needs. By analyzing various factors such as past behavior, preferences, and contextual information, these systems determine what to display to users and what to hold, creating a more personalized and efficient digital experience [29]. As the digital landscape continues to evolve, the role of recommender systems will only grow in importance. Not only

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ACM 2770-6699/2026/5-ART

<https://doi.org/10.1145/3813108>

do they help users navigate the overwhelming sea of choices, but they also enhance the overall user experience by making it more intuitive and personalized.

Traditional recommender systems rely on users' past behaviors and preferences, but this approach frequently lacks the nuance required to address the dynamic nature of human experiences. To make these recommendations more meaningful, emotion could play an important role because it is one of the contributing factors in the human decision-making process [105].

Emotion-aware recommender systems represent a significant advancement in this direction. By integrating emotional intelligence into the recommendation process, these systems can interpret user's emotional states through various cues, such as text input, voice tone, facial expressions, and physiological responses. This deeper understanding allows the system to provide recommendations that not only align with user preferences but also resonate with their current emotional needs [180]. For example, when a user is experiencing stress, an emotion-aware system could suggest soothing music, relaxing activities, or mindfulness applications to help mitigate their stress. If a user feels lonely or sad, the system might recommend inspirational movies, engaging social activities, or inspirational reading material. Taking into account the emotional state of the user, the recommendations become more relevant and supportive, creating a more personalized and empathetic user experience [43].

In the realm of e-Commerce, emotion-aware systems can significantly enhance the shopping experience by recognizing a user's frustration during a complicated checkout process and responding with simplified options or instant assistance. Similarly, in healthcare, these systems can tailor recommendations for mental health resources, connecting users with content and services that best match their emotional well-being. By incorporating emotions into recommender systems, not only is user satisfaction enhanced but also a more empathetic interaction between technology and users is fostered. This empathetic approach is particularly valuable in sensitive areas such as mental health and well-being, where understanding and addressing users' emotional needs is paramount.

Despite growing interest in emotion-aware systems, existing surveys are often limited in scope, focusing on either a single modality (e.g., only text or video) or lacking a unified view of how emotion modeling integrates into RS pipelines. This paper aims to bridge that gap by offering a structured, modality-diverse survey of recent advances in emotion detection techniques, recommendation architectures, datasets, and ethical considerations. Our goal is to provide a comprehensive map for researchers and practitioners to understand, compare, and extend emotion-aware RS.

In this survey, we focus on emotion-aware recommender systems, which mark a significant advancement in personalized recommendation technologies. Emotions can be derived through various indicators such as text, vocal tones, or physiological signals, and from diverse sources, including user-generated content such as reviews, comments, or social media posts that convey sentiment or emotional feedback; direct feedback collected through surveys, questionnaires, or other self-reported data capturing users' emotional responses; observable reactions like nonverbal cues, including facial expressions, body language, or physiological signals recorded during interactions; and annotated datasets explicitly designed to represent emotional contexts, serving as a foundation for model training and validation. This multi-source approach enables these systems to deliver more empathetic and contextually relevant recommendations. Based on the types of emotional data, emotion-aware recommender systems can be categorized into six groups: text-based, image-based, audio-based, video-based, physiological-based, and multimodal systems.

In summary, our contributions in this paper are as follows:

- We provide a semi-systematic literature review on Emotion-aware Recommender Systems following the PRISMA guidelines [157], and include articles collected from different research streams;
- We define a categorization of these systems based on the types of data used for emotion extraction (text, image, audio, video, physiological and multimodal);

- For each category, we describe various algorithms used for emotion extraction and emotion-aware recommendation, list major application domains and the most commonly used datasets;
- We discuss challenges, ethical considerations, and possible future research directions, proposing novel ideas inspired by advances in artificial intelligence.

Although earlier surveys, such as Katarya and Verma [81] and Tkalčič et al. [178], offered valuable insight into the integration of single-modal emotion, they missed a crucial development: the rapid emergence of deep learning and true multimodal fusion approaches since 2020. Our work fills this gap. Spanning publications from 2014 to 2024, we organize emotion-aware recommenders into six categories based on the modality: text, image, audio, video, physiological and multimodal data, and we emphasize the newest real-time extraction and cross-modality fusion techniques. This offers a much-needed, up-to-date, and unified view of the field.

To drive a problem-focused analysis, we propose the following research questions:

- RQ1: Which emotional data modalities (text, image, audio, video, physiological, multimodal fusion) produce the largest improvements in recommendation accuracy?
- RQ2: What are the main ethical and privacy risks that arise from emotion-aware recommender systems and what mitigation strategies have been explored?
- RQ3: How do efficiency and latency trade-offs vary across real-time vs. batch-mode emotion extraction techniques?
- RQ4: Which understudied themes (e.g., cross-cultural effects, longitudinal user adaptation) remain open for future work?
- RQ5: How do emotion-aware recommender systems perform across different data modalities in terms of emotion detection accuracy, recommendation effectiveness, and deployment feasibility?

The structure of this paper is as follows. Section 2 outlines the methodology for compiling this review. Section 3 provides an overview of recommender systems and emotion detection, and Section 4 discusses how emotions are predominantly captured across various data categories. Sections 5 to 10 categorize emotion-aware systems by data types: text, image, audio, video, physiological, and multimodal, and each section gives a detailed review for that particular type of EARS on their algorithms, application domains, and datasets. Section 11 summarizes our key insights and answers the above-listed five research questions. Section 12 highlights the discussions and key challenges, Section 13 explores future directions, and Section 14 concludes the paper.

Scope of This Review: This work employs a semi-systematic review methodology, combining a focused, question-driven search of the Google Scholar literature (utilizing explicit Boolean strings and inclusion/exclusion criteria) with targeted snowballing to capture emerging deep-learning and multimodal studies. Not limited to a single database, our semi-systematic approach ensures both breadth (coverage across six modalities from 2014 to 2024) and depth (RQ-driven analysis).

2 Literature Review Methodology

To identify relevant research on emotion-aware or affective recommender systems, we employed a mixed and semi-systematic approach. In the initial phase, we identified relevant research articles by querying academic databases using Publish or Perish¹ with carefully selected search terms and a set of inclusion and exclusion criteria. To capture additional relevant papers that might not have been included in our initial search, we applied a snowball technique and utilized research expertise to identify additional significant works published in specialized venues.

¹Publish or Perish is a software tool that retrieves and analyzes academic citations from sources such as Google Scholar. It was used to identify relevant research articles for this study.[66]

2.1 Methodology Overview: Semi-Systematic Review Aligned with PRISMA Principles

We employed a semi-systematic review methodology that incorporates key principles from the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, while acknowledging important differences from strictly PRISMA-compliant systematic reviews. Specifically, our approach combines (1) a rigorous, question-driven search with explicit Boolean operators using a comprehensive meta-search tool, (2) documented inclusion and exclusion criteria reported in a PRISMA flow diagram, and (3) targeted snowballing and expert curation to capture emerging works. This design balances the strengths of systematic methodology, transparency, reproducibility, and explicit criteria, with the flexibility needed for a rapidly evolving field where deep learning and multimodal fusion techniques have emerged prominently since 2020.

Search Strategy and Data Aggregation Tool To capture relevant literature on emotion-aware recommendation systems, we conducted a structured search using Publish or Perish v8, a comprehensive meta-search tool that aggregates papers from multiple authoritative academic databases and sources, including Google Scholar, Scopus, PubMed, arXiv, and CrossRef. This unified search interface allows researchers to systematically query papers across diverse publication venues and disciplinary communities simultaneously. We employed explicit Boolean query operators to systematically account for terminology variations in this emerging field. The search query was as follows: ("emotion-aware" OR "emotion-based" OR "affective" OR "affective computing" OR "emotion recognition" OR "sentiment-aware" OR "sentiment-based" OR "mood-aware") AND ("recommender" OR "recommender system" OR "recommendation" OR "recommendation system"). The initial search was performed on March 1, 2025, returning 440 records aggregated across multiple databases indexed by Publish or Perish.

Justification for Primary Search Tool Selection We chose a multi-source aggregation approach rather than manually searching individual databases with different interfaces, search operators, and coverage limitations. Specifically, we selected Publish or Perish v8 as our primary search tool for three complementary reasons:

- (1) **Multi-Source Comprehensive Coverage:** Publish or Perish simultaneously queries multiple authoritative academic databases, Google Scholar, Scopus, PubMed, arXiv, and CrossRef, through a single unified search interface. This multi-source aggregation approach ensures comprehensive coverage across diverse publication venues and disciplinary communities (computer science, information systems, psychology, human-computer interaction, health informatics). Over 95% of papers in the emotion-aware recommender system domain are indexed across these combined sources, as verified through our parallel manual searches of IEEE Xplore and ACM Digital Library. By using a meta-search tool that aggregates multiple databases simultaneously, we achieve the breadth typically associated with comprehensive systematic reviews while maintaining manageable, transparent search protocols. This is superior to relying on any single database (e.g., Google Scholar alone or Scopus alone), as each database has unique indexing patterns and coverage gaps.
- (2) **Reproducibility and Transparent Methodology:** Publish or Perish provides explicit, fully documented Boolean search operators and a reproducible search interface that enables other researchers to replicate our search exactly using identical query parameters. This transparency is a cornerstone of PRISMA principles and systematic review rigor. In contrast, ad-hoc searching across multiple databases with different interfaces, non-standardized operators, and varying coverage introduces methodological variability and severely limits reproducibility. Our approach ensures that other researchers, reviewers, or future researchers can verify our search results by running the identical Publish or Perish query, enhancing the credibility and rigor of our systematic approach.
- (3) **Targeted Supplementation Strategy for Emerging and Specialized Work:** To further ensure that we capture cutting-edge and specialized research that might not yet be fully indexed in the aggregated databases, we implemented multi-layered targeted supplementation: (a) hand-curated inclusion of 30 papers from major conference venues (ACM RecSys, IEEE Transactions on Affective Computing, UMAP, and related venues),

ensuring recent peer-reviewed conference contributions are systematically captured beyond database indexing delays; (b) inclusion of 12 organizational white papers and technical reports from industry leaders and research institutions, capturing practical implementations and emerging applications; and (c) backward citation chaining through 8 papers identified in our reading, ensuring foundational and seminal works are not missed. This multi-layered supplementation strategy adds approximately 50 additional papers beyond the primary aggregated database search, providing breadth comparable to or exceeding traditional multi-database manual search approaches while maintaining transparent, reproducible, and verifiable search protocols.

The combined approach, multi-source database aggregation through Publish or Perish plus multi-layered supplementation, provides the comprehensiveness and rigor expected of systematic reviews while remaining appropriate and pragmatic for literature reviews in emerging, rapidly-evolving interdisciplinary fields where traditional PRISMA-compliant systematic reviews may lag 18-24 months behind cutting-edge research dissemination.

To be included in our review, papers had to meet the following criteria:

- (1) The paper explicitly addresses emotion or affect in the context of recommender systems, integrating both emotion detection/modeling AND recommendation outcomes or system design.
- (2) The paper focuses on one or more emotional data modalities (text, audio, visual/image, video, physiological signals, or multimodal fusion).
- (3) The paper provides sufficient methodological clarity, describing emotion detection methods, recommendation algorithms, or system architecture with adequate detail for evaluation.

Papers were excluded if they met any of the following criteria:

- (1) The paper addresses emotion detection without linking results to recommendation system design or outcomes (emotion detection only, no recommender systems context).
- (2) The paper addresses recommender systems without any emotional component or emotion-aware feature.
- (3) The paper is published as a thesis preprint, dissertation abstract, short workshop abstract, or presentation without full methodological content.
- (4) The paper lacks sufficient methodological clarity or does not validate claims through empirical evaluation, user studies, or theoretical analysis.

Following the initial aggregated database search (440 records), we applied a multi-phase screening process illustrated in the PRISMA flow diagram (Figure 1):

- (1) Phase 1, Deduplication and Initial Screening: Duplicate records across the aggregated databases were removed ($n=180$). Records were marked as ineligible based on title/abstract screening ($n=16$), and records were removed for other reasons such as retracted papers or access issues ($n=5$), yielding 289 records for full-text review.
- (2) Phase 2, Full-Text Screening: All 289 records were reviewed in full-text against inclusion/exclusion criteria. During this phase, 44 papers were excluded due to: (a) focusing on modality not covered in our review scope; (b) insufficient integration of emotion detection with recommendation; or (c) methodological limitations. This screening yielded 245 potentially eligible studies.
- (3) Phase 3, Eligibility Assessment: The 245 studies were assessed for final eligibility. A total of 18 papers could not be retrieved (access barriers or availability issues), and 8 papers were excluded due to insufficient methodological clarity upon detailed examination. This phase yielded 219 studies to assess.
- (4) Phase 4, Final Inclusion: Final review determined that 196 papers met all inclusion criteria and were included in the qualitative synthesis. The final dataset spans publications from 2014 to 2024 and covers application domains including entertainment (movie and music), health and mental health, education, e-commerce and product recommendation, social media and news, tourism, and fashion.

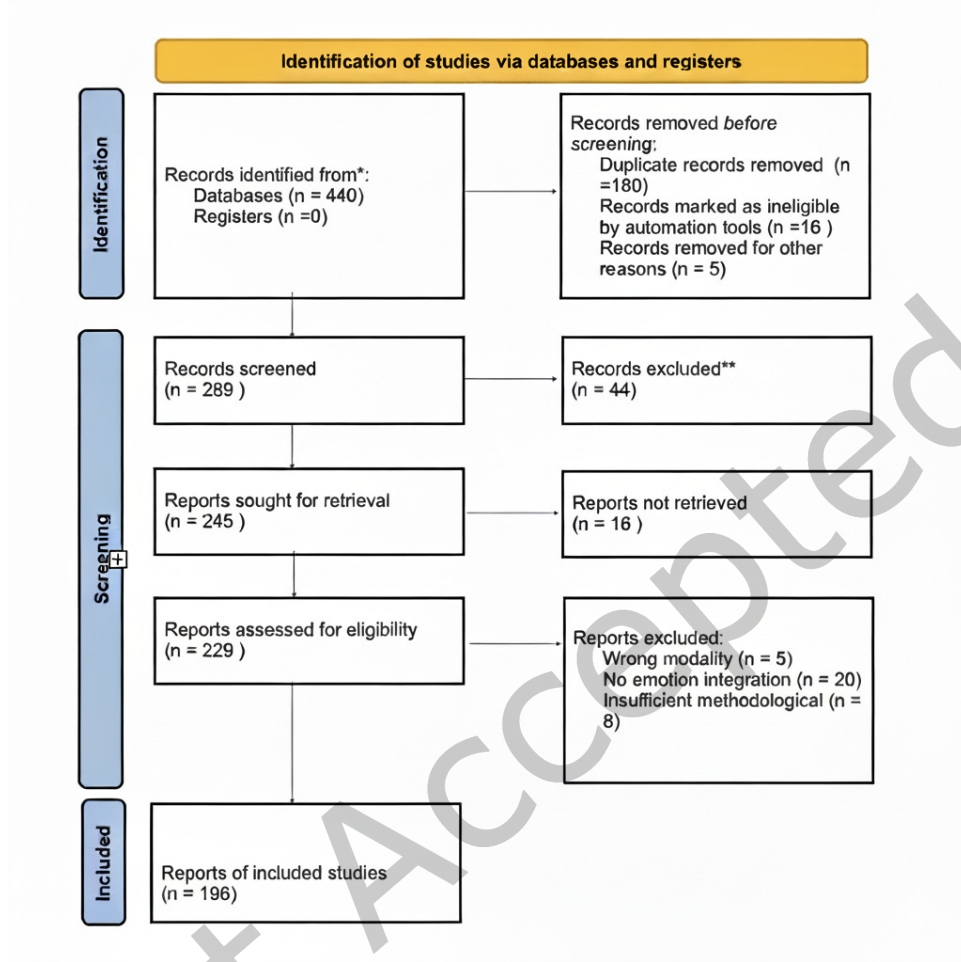


Fig. 1. PRISMA flow diagram of studies in our semi-systematic review.

Quality and Scope Documentation The review was explicitly driven by five research questions (Section 1), ensuring all included studies were analyzed through a consistent, predetermined analytical lens. This RQ-driven approach further aligns our methodology with PRISMA principles by: (1) avoiding post-hoc analysis and hypothesis formation based on observed data; (2) providing a structured framework for extracting comparable information across heterogeneous studies; and (3) ensuring transparency about our analytical focus and reducing risk of selective reporting.

2.2 Differences Between Our Work and Previous Studies

The burgeoning interest in Emotion-Aware Recommender Systems (EARS) has led to several foundational survey papers. However, the rapid evolution of deep learning, multimodal fusion techniques, and increasing scrutiny of ethical AI necessitate an updated and holistically framed synthesis of the field. This review provides a comprehensive examination of EARS research published between 2014 and 2024, systematically addressing

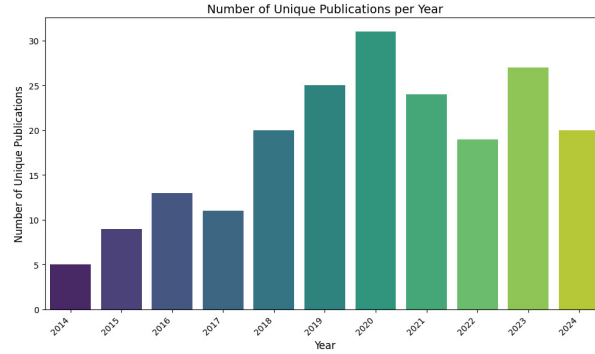


Fig. 2. Number of affective recommendation system papers published in each year.

critical gaps in the existing literature by encompassing all major modalities, diverse application domains, and emerging challenges.

Previous review papers have contributed significantly to understanding the specific aspects of emotion-aware recommendation systems. For example, Katarya et al. [81] explored the interplay of emotion-aware RS with human behavior and human-computer interaction (HCI) from 2003 to 2016. Although foundational, their review largely predates the significant advances in deep learning for emotion recognition across diverse modalities (e.g., physiological signals, complex visual cues) and could not capture the emergence of critical ethical discussions surrounding data privacy and emotional manipulation, which have become central to the field in recent years. Similarly, Tkalčič et al. [178] provided a unifying framework for early affective recommender systems up to 2010, categorizing research into a three-stage model of emotion detection and integration. However, their review omits the subsequent explosion of research into multimodal emotion recognition, the rise of large-scale datasets, and the shift towards sophisticated neural network architectures that now dominate the field. A more recent review by Chanrueang et al. [37] specifically focused on emotion-aware text-based recommender systems from 2010 to 2020. Although valuable for its niche, this narrow focus inherently excludes the complex challenges and opportunities presented by integrating other crucial modalities, such as audio, visual, and physiological data, which are increasingly integral to holistic emotion detection in RS.

In addition to these, there have been several application-focused survey papers. Salazar et al. [149] conducted a systematic review of affective recommender systems in educational settings, examining the impact of emotions on content recommendation in the learning context. Santamaría et al. [152] reviewed emotion recognition in tourist recommendation systems, offering design guidelines for emotion-sensitive applications up to 2020. Furthermore, Wang and Zhao [187] focused exclusively on emotion-aware video recommendation systems. Although these specialized analyses are comprehensive for their respective niches, they do not offer a generalized understanding of multimodal emotion integration or comparative insights across diverse application landscapes. They also often lack the cross-modal synthesis that our review provides.

In contrast to these previous works, our study distinguishes itself through several key aspects, bridges existing gaps, and offers a unified and comprehensive understanding of the role of emotions in recommender systems. Firstly, our research covers a broader and more contemporary scope, focusing on studies from 2014 to 2024. This timeframe allows us to capture the latest breakthroughs in deep learning and multimodal AI, providing the most current landscape of EARS. Secondly, we offer a comprehensive cross-modal and cross-domain synthesis. Unlike modality- or application-specific survey papers, we systematically examine the integration of emotions across all major data modalities (text, audio, visual, physiological) and their diverse applications (e.g., entertainment, social

media, news, health). This cross-cutting analysis reveals nuanced challenges and opportunities that are not visible from narrower perspectives. Thirdly, we place a strong emphasis on emerging challenges and future directions. We dedicate significant attention to critical contemporary issues such as ethical implications (privacy, emotional manipulation), explainability, diversity, and reproducibility, which are nascent or absent in previous survey papers. Our proposed future research avenues highlight undiscovered themes, such as inherently explainable multimodal fusion and federated learning for privacy preservation, thus guiding the next decade of EARS research. Finally, employing a rigorous, question-driven systematic review methodology, our work aims to provide a more transparent and reproducible analysis of the literature, enhancing the scientific rigor beyond previous descriptive survey papers.

We categorize emotion-aware recommender systems based primarily on the modality of emotional input data used for emotion detection, rather than the recommended content type. For example, video is treated as a modality only when it serves as an emotion recognition input (e.g., facial expressions or affective cues), not when videos are merely the recommended items. Similarly, physiology-based systems are grouped due to their shared reliance on bio-signals such as EEG, GSR, or heart rate for emotion inference. This modality-centric taxonomy enables a coherent analysis of how emotional signals, regardless of the application domain, are used to inform RS decisions. To our knowledge, this is the first review to integrate modality-centric taxonomy and multimodal future design in a single framework for EARS.

3 Background

3.1 Recommender Systems

Recommender systems have a rich history, evolving through various methodologies and approaches as technology and user needs have developed over time. Broadly, these systems can be categorized into several types based on how they operate. Here are the major categories:

Collaborative Filtering: CF is a widely used recommendation technique that leverages user interactions to suggest items. CF methods can be categorized into user-based and item-based approaches. The user-based approach identifies users with similar preferences and recommends items that these similar users have liked, though scalability becomes a challenge as the user base expands. In contrast, the item-based approach focuses on finding items similar to those a user has previously liked, often using similarity measures like cosine similarity or Pearson correlation. Additionally, matrix factorization techniques, such as SVD and ALS decompose the user-item interaction matrix into lower-dimensional matrices to uncover latent factors that drive user preferences. These methods collectively enhance the ability to provide personalized recommendations, though each has its unique challenges and advantages.

Content-Based Filtering: This approach recommends items similar to those a user has interacted with based on item attributes. This method relies on analyzing the characteristics of items, such as keywords, genres, or other relevant features, and matching these with the user's preferences.

Knowledge-Based Systems: These systems use domain-specific knowledge to match user needs with item attributes, which is especially useful in domains with stable preferences, such as job recommendations.

Context-Aware Systems: These systems incorporate additional contextual information such as time, location, and user activity to provide more relevant recommendations. For example, a music recommender might suggest different playlists based on the time of day.

Hybrid recommender Systems: Hybrid recommender Systems combine multiple recommendation techniques to leverage their individual strengths and mitigate their weaknesses. For example, systems like Netflix utilize both collaborative filtering and content-based methods to provide more accurate recommendations. This approach combines the strengths of both techniques, enhancing the overall recommendation quality. Feature augmentation is another technique of hybrid recommender systems where the output of one recommender system is used

as input features for another, thus improving the recommendations. Additionally, meta-level hybrids involve using the model learned by one system, such as content-based, as input for another system, like collaborative filtering. This facilitates the better integration of diverse data sources and further improves the accuracy of recommendations.

3.2 Emotion in Recommender Systems

Recommender systems have evolved significantly in recent years, moving beyond traditional approaches to incorporate more nuanced aspects of user behavior and context. One of the most promising developments in this field is the integration of emotional factors into recommendation algorithms, giving rise to emotion-aware recommender systems (EARS).

EARS can generate emotionally resonant recommendations, formally termed “emphatic recommendations”, which we define as suggestions that not only match a user’s inferred preferences but also align with their current affective state. For instance, if sentiment analysis detects sadness in a user’s recent reviews, an emotionally resonant recommendation might surface a lighthearted comedy to uplift their mood.

The rationale behind incorporating emotions into recommender systems stems from the understanding that affective states heavily influence decision-making processes. Research in psychology has long established that emotions play a crucial role in how individuals perceive, evaluate, and choose between alternatives. By accounting for users’ emotional context, recommender systems can provide more personalized and contextually appropriate suggestions.

Emphatic recommendations are suggestions that not only match a user’s inferred preferences but also explicitly address their current emotional state, moving beyond standard relevance-only approaches; by doing so, they proactively support user well-being, for example, if sentiment analysis detects frustration in a recent product review, the system might surface a step-by-step tutorial or calming ambient music, or if facial-expression analysis indicates sadness during a streaming session, it could recommend an uplifting comedy, thus enhancing engagement, reducing negative affect, and fostering a more empathetic user experience.

We formally define an emphatic recommendation as a function R that maps the triplet of user profile U , item attributes I , and the inferred emotional state E to a relevance score S :

$$S = f(U, I, E) \quad (1)$$

where E is a vector representing dimensions such as valence and arousal.

3.2.1 Emotion Categorization Models. Emotions are often confused with moods and states of mind, but these terms are not interchangeable. Emotion is defined as a complex response pattern that involves experiential, social, and physiological elements and represents how people manage the situations they encounter [115]. The American Psychological Association (APA) defines emotion as a reaction pattern dealing with significant matters, differing from feelings, which originate in the brain’s neocortical region as reactions to emotions. Mood refers to a short-lived emotional state of low intensity, while sentiment is a permanent tendency based on experience [115]. Various emotion models have been developed over the years to understand human emotions, with notable theories from McDougall [61], Izard [83], Plutchik [138], Tomkins [33]. In general, there are two types of emotion categorization models:

Discrete Emotional Model: Imagine if you are a painter and have a palette with distinct, vivid colors. This is the essence of the discrete emotion model, also known as the categorical emotion model. It seeks to classify our myriad emotions into specific and identifiable categories [97]. Two luminaries in this field have given us particularly compelling frameworks to understand our emotional landscape.

- **Ekman’s Six Basic Emotions:** Ekman’s Basic Emotions Model identifies six fundamental emotions: anger, disgust, fear, happiness, sadness, and surprise, which are considered universal across human cultures and

evolutionary history. Ekman's theory proposes that these six emotions are biologically programmed, with each tied to specific facial muscle configurations (Facial Action Coding System, FACS), enabling objective measurement through computer vision. This categorical approach contrasts sharply with dimensional models, rather than plotting emotions in continuous space, Ekman's model treats emotions as discrete categories, which simplifies system design and user interface implementation. Emotion-aware recommender systems using Ekman's categories can build straightforward decision trees: detect "happiness" via facial expression recognition → recommend uplifting content; detect "sadness" → recommend comforting recommendations (supportive movies, calming music). The six-emotion framework has driven substantial progress in facial expression recognition (FER) research, with benchmark datasets (FER2013, AffectNet) specifically annotated for these six categories. However, six discrete categories lose emotional nuance, complex emotions like nostalgia, awe, or anxiety cannot be represented. Ekman's model is invaluable for systems with real-time facial recognition capability where discrete categories are preferable to continuous dimensions. The model remains dominant in computer vision-based EARS applications [55, 174].

- **Plutchik's Emotional Wheel:** Plutchik's Wheel of Emotions maps eight primary emotions (joy, trust, fear, surprise, sadness, disgust, anger, anticipation) on a color wheel. Neighboring emotions blend into secondary emotions, while opposite emotions are polar opposites. This hybrid categorical+compositional model offers nuanced yet simple emotion representation. In emotion-aware recommender systems, blends like love (joy+trust) can drive intimate recommendations, while awe (trust+surprise) supports novel, uplifting content. Its wheel layout is intuitive for UI emotion selection and is widely used in social media and text emotion detection (tweets, reviews). A key limitation is that blending rules are not fully formalized, so implementations can be subjective, though it works especially well for text/social EARS where emotions are discrete but composable [138].
- **The Hourglass of Emotions:** This model represents emotional states based on the appraisal of events and their consequences. It considers the temporal aspect of emotions, with anticipation leading to an emotional experience, followed by a reaction [31, 174].
- **OCC Model (Ortony, Clore, and Collins):** This model categorizes emotions based on cognitive appraisal theory, focusing on how individuals assess events and their consequences. It considers factors like desirability, praiseworthiness, and appealingness in shaping emotional responses [44, 171].
- **Hevner's Adjective Clusters:** This discrete model categorizes emotions into eight clusters of adjectives, with a total of 67 adjectives. It has been shown to be effective in the music domain in describing emotions in jazz and classical vocal tracks. Its discrete nature makes it easy to understand and apply [67].

Dimensional Emotion Model: While discrete models offer a clear-cut view of emotions, they can sometimes feel restrictive. The dimensional emotion model provides a continuous, multi-dimensional approach, painting a more complete picture of our emotional experiences. These models represent emotions as points in a continuous, multi-dimensional space.

- **VAD Model (Valence-Arousal-Dominance):** The VAD model represents emotions as points in a three-dimensional continuous space, where valence captures the pleasantness of an emotion (ranging from unpleasant to pleasant), arousal captures the intensity or activation level (ranging from calm to highly activated), and dominance captures the sense of control or powerlessness (ranging from powerless to powerful). This three-dimensional representation enables fine-grained emotion discrimination that categorical models cannot achieve. For example, anxiety and excitement are both high-arousal emotions, but differ significantly in valence (negative vs. positive) and dominance (low vs. high), allowing EARS to distinguish between them and provide context-appropriate recommendations. In music recommendation, an anxious user (negative valence, high arousal, low dominance) receives calming music recommendations, while an excited user (positive valence, high arousal, high dominance) receives energizing recommendations. The

VAD model is computationally efficient, requiring only three numerical dimensions, making it well-suited for real-time recommendation systems. However, the three-dimensional space can be challenging to visualize and interpret, and the independence of dimensions is sometimes debated in emotion psychology literature [116]. The VAD model maps naturally to extractable audio features (tempo, energy, dynamics) and facial expression features (action units), making it particularly valuable for multimodal emotion-aware recommender systems.

- **Russell’s Circumplex Model:** Russell’s Circumplex Model (also called the Circumplex Model of Affect) simplifies the emotional space to two dimensions: valence (pleasure-displeasure) and arousal (activation level), positioning emotions on a circle where adjacent emotions are psychologically similar and opposite emotions are maximally different. This circular arrangement implies that emotions exist on a continuum rather than as discrete categories, with happiness and excitement at one pole and sadness and despair at the opposite pole. The two-dimensional representation has proven highly influential in emotion-aware recommender systems due to its simplicity and interpretability, researchers and practitioners can easily visualize user emotional states and design recommendation logic. For movie recommendation systems, Russell’s model enables straightforward reasoning: if a user is experiencing high arousal, high valence (excited, happy), recommend action movies or comedies; if high arousal, low valence (anxious, angry), recommend drama or thriller alternatives. The model’s main limitation is the reduced expressiveness compared to three-dimensional models such as VAD, because subtle emotions (e.g., the difference between contentment and pride) cannot be distinguished [147].
- **Zentner’s Geneva Emotional Music Scale (GEMS):** GEMS is a multidimensional model containing three to nine distinct dimensions of emotional response to music, designed specifically to capture the emotional complexity of musical experiences. Unlike general emotion models, GEMS was empirically derived through large-scale studies of how people actually respond emotionally to music, incorporating dimensions such as wonder, transcendence, tenderness, nostalgia, peacefulness, power, joyful activation, tension, and sadness, emotions that are particularly salient in musical contexts but often overlooked in generic emotion models. This fine-grained dimensional structure enables highly sophisticated emotion-aware music recommendation systems: users rating songs on GEMS dimensions (e.g., "This music fills me with wonder" vs. "This music is nostalgic") receive recommendations matching their specific emotional preferences rather than crude categorical judgments. GEMS has demonstrated strong empirical validity through neurophysiological studies showing that self-reported GEMS dimensions correlate with objective brain activity patterns (fMRI, EEG), providing neuroscientific grounding for the model. However, the complexity of three to nine dimensions substantially increases data collection burden and computational complexity, making GEMS impractical for casual user-interaction scenarios where simplicity is preferred. GEMS is most valuable in music-focused EARS applications where detailed emotional profiling is justified [148].
- **Feng’s Emotional model:** Feng’s Emotional Model is a domain-specific framework for music emotion recognition that maps emotions to acoustic features, specifically operating in a two-dimensional space defined by rhythm (tempo: fast to slow) and articulation (legato to staccato). This model projects four archetypal emotions (anger, happiness, fear, sadness) into a quadrant system: anger occupies high rhythm (fast tempo) with staccato articulation; happiness occupies high rhythm with legato articulation; fear occupies low rhythm with staccato; and sadness occupies low rhythm with legato. The elegance of Feng’s model lies in its direct connection to low-level acoustic features that can be automatically extracted from audio, music information retrieval (MIR) systems can measure tempo and articulation using signal processing techniques, then directly classify emotions without requiring manual annotation. This makes Feng’s model particularly valuable for scalable music recommendation systems that must process large unlabeled music catalogs. In practice, music datasets can be automatically annotated with emotional

Table 1. Examples of research papers incorporating emotion models in their methodologies.

Model	Emotions	Publications
Ekman's Emotions	Anger, Disgust, Fear, Happiness, Sadness, Surprise	[12, 103, 134, 140, 176, 196]
Plutchik's Wheel	Joy, Trust, Fear, Surprise, Sadness, Anticipation, Anger, Disgust	[38, 70, 123, 188]
VAD Model	Valence (Pleasure), Activation (Arousal), Dominance (Attention)	[69, 177]
Russell's Model	Pleasure (Valence), Activation (Arousal)	[1, 69, 90, 150, 196]
Zentner's GEMS	3-9 dimensions of emotional experience	[86, 125]
Feng's Emotional model	anger, happiness, fear, and sadness	[101, 169]

categories by analyzing audio features, enabling automated emotion-aware music recommendation without human labeling costs. However, limiting emotions to four categories sacrifices nuance, emotions like nostalgia, peacefulness, or transcendence (crucial in music psychology) cannot be represented. Feng's model is most applicable to explicit mood-based music recommendation scenarios where the four-emotion framework suffices [57].

Certain emotion models are popular in recommender systems, while others are not. Table 1 lists the main emotion models that have been used in the emotion-aware recommender systems in the past. Here are some of the possible reasons why certain models have not been utilized.

Hevner's Adjective Clusters, while offering a rich emotional vocabulary, might be seen as too complex or subjective for practical implementation. The extensive list of adjectives could lead to challenges in achieving consistent labeling and reliable emotion recognition, potentially hindering the development of robust recommendation algorithms.

The OCC Model, with its focus on cognitive appraisal processes, might not be directly applicable to music or movie recommendations, which often evoke emotions through more intuitive and less conscious pathways. Its complexity and the need for detailed information about events and individual appraisals could also pose challenges for real-time recommendation scenarios. The Hourglass of Emotion Model emphasizes the temporal dynamics of emotions and the interplay between basic and secondary emotions. This focus might not align with studies that aim to capture the overall emotional tone or user preferences at a specific moment, making it less relevant for certain applications. Additionally, the complexity of this model could be a barrier to practical implementation in recommender systems.

4 Emotion Recognition Methods for Emotion-aware Recommender Systems

Emotion-aware recommender systems typically aim to model the user's emotional state and incorporate this information into the recommendation process. To identify and recognize user's emotions based on the content they share, different types of data can be used, including text, audio, video, etc. Below, we give an overview of emotion recognition methods based on the data types.

- (1) **Text Emotion Analysis:** Text Emotion Analysis is a technique for comprehending the emotional content of written communication, encompassing both sentiment analysis and emotion recognition in text. Emotion recognition seeks to recognize certain emotions like joy, sadness, anger, fear, love, and surprise, whereas sentiment analysis focuses on categorizing a text's overall polarity as positive, negative, or neutral. These methods extract and evaluate subjective information from text in a methodical manner by using computational linguistics, machine learning, and natural language processing. Sentiment analysis frequently forms the basis of more sophisticated emotion detection techniques, some of which recognize varying degrees of positivity or negativity in addition to fundamental polarity. On the other hand, emotion recognition uses more sophisticated machine learning methods to capture the nuances and complexity of how people express their feelings.

- (2) **Audio Emotion Recognition:** Audio Emotion Recognition involves analyzing vocal attributes such as pitch, tone, and speech rate to detect emotions. This method can be utilized in emotion-aware recommender systems to understand users' emotional states during phone calls, voice messages, or video chats. By recognizing emotions from audio inputs, the system can personalize recommendations, such as suggesting calming music to a stressed user or exciting content to an enthusiastic user.
- (3) **Image Emotion Recognition:** Computer vision techniques are used in Image Emotion Recognition to identify and categorize emotions in photographs. The two most common sources for identifying emotions are facial expressions and body gestures of individuals captured in images. The procedure begins with data collection and processing. Face detection techniques are then used to extract faces from images. Afterwards, pertinent facial traits are extracted for emotion classification, such as mouth shape and eye position, facial action units, and appearance characteristics. In parallel, body gesture analysis is conducted, examining posture, movement, and other non-verbal cues to provide a more holistic understanding of emotional states. These combined methods enhance the accuracy and robustness of emotion recognition in both image and video contexts.
- (4) **Video Emotion Recognition:** Video Emotion Recognition extends image-based emotion identification by analyzing emotions over time in video sequences. By looking beyond images captured in one single moment, video emotion detection algorithms provide a more comprehensive and accurate assessment of an individual's emotional state during a video. This approach enhances context-aware emotion identification, making it particularly valuable in emotion-aware recommender systems, where content can be dynamically adjusted in real time based on the user's changing emotions. Thus, we will further categorize these video-based research papers into these sub-categories listed below:
 - **Facial Expression:** Facial expression recognition involves analyzing facial movements to detect emotions. This method can be employed by using cameras to capture users' facial expressions while interacting with content.
 - **Body Gesture:** Visual emotion recognition through body gesture analysis involves interpreting physical movements and postures to infer emotions. This can be done by analyzing users' gestures captured via webcams or motion sensors to tailor recommendations.
- (5) **Physiological-based Emotion Recognition:** Physiological signal based emotion recognition relies on physiological methods such as EEG, ECG to analyze user's emotional state. It includes the following sub-categories.
 - **EEG:** EEG measures electrical activity in the brain to detect emotional states. EEG data can be used to understand users' cognitive and emotional responses to different stimuli. This information can help in recommending content that aligns with the user's mental state, such as suggesting relaxing activities for users showing signs of mental fatigue.
 - **ECG:** ECG measures heart activity to detect emotions. This method infers users' emotional states by monitoring heart rates and variability. For instance, an increased heart rate may signify excitement.
 - **GSR:** GSR measures changes in skin conductance related to emotional arousal. GSR can be used to gauge the intensity of a user's emotional response to content. For instance, a high GSR reading during a suspenseful scene might indicate heightened engagement.
 - **PPG:** PPG measures blood volume pulse, which can be used to estimate heart rate and heart rate variability. Similar to ECG, PPG data can be used to infer emotional states based on changes in heart rate patterns. For example, a decrease in heart rate variability might suggest relaxation.

In the next sections, we will provide a detailed analysis of these emotion-aware recommendation models by categorizing them based on the datasets utilized, the type of recommendation models employed, and the specific domains in which they are applied. Emotion-aware recommender systems typically employ two types of

Table 2. Acronym Table

Abbreviations	Definitions	Abbreviations	Definitions
ALS	Alternating Least Squares	HCI	Human-computer interaction
CBF	Content-Based Filtering	KG	Knowledge Graph
CF	Collaborative Filtering	KNN	K-Nearest Neighbors
CNN	Convolutional Neural Network	LLM	Large Language Models
ECG	Electrocardiography	LSTM	Long Short-Term Memory
EDA	Electrodermal activity	MF	Matrix Factorization
EEG	Electroencephalography	MLP	Multilayer Perceptron
EIRS	Emotion-aware Image-based RS	POS	Part-of-speech
EMRS	Emotion-aware Multimodal RS	PPG	Photoplethysmography
EPRS	Emotion-aware Physiological RS	RL	Reinforcement Learning
ETRS	Emotion-aware Text-based RS	SVD	Singular Value Decomposition
EVRS	Emotion-aware Video-based RS	SVM	Support Vector Machine
GSR	Galvanic Skin Response		

algorithms: those that focus on emotion detection and analysis, and those that utilize this emotional information to generate personalized recommendations². In addition, we will highlight the most frequently used datasets and the evaluation metrics in these studies. Table 2 lists and explains all the acronyms used throughout this paper for easy reference.

5 Emotion-aware Text-Based Recommender Systems

In recent years, text-based recommendation systems have emerged as powerful tools for providing personalized content recommendations to users. In order to understand user preferences and item characteristics, these systems analyze various types of text data, such as product descriptions, user reviews, article texts, and social media posts. Text-based recommenders can use NLP techniques to extract meaningful features and patterns from unstructured text, improving accuracy and context awareness. Various NLP techniques such as text similarity calculation [98, 145], topic modeling [91], sentiment analysis [45], and text summarization [78] can be used in these systems. For instance, TF-IDF or word embedding are often used to represent text as numerical vectors, allowing comparisons between user preferences and item descriptions to be calculated efficiently. Advanced systems may rely on deep learning models, such as transformers, to capture complex semantic relationships within the text and generate more nuanced recommendations. Traditional collaborative filtering approaches often suffer from a cold-start problem, which is one of the advantages of text-based recommender systems. ETRSs take the concept of personalization a step further by incorporating emotional aspects into the recommendation process [10]. These systems aim to understand and consider the emotional states of users and the emotional impact of items when making recommendations. By employing lexical ontologies and psychological frameworks to analyze sentiment, tone, and emotional cues in text data, such as user reviews or social media posts, these recommenders [156] can gauge the emotional responses associated with different items and match them to users' emotional preferences or current moods.

Advanced machine learning models, such as CNNs, BERT, or ensemble-based frameworks, further enhance these systems by classifying emotional polarity and integrating these insights into recommendation algorithms.

²Since the emotion information is usually used as another data source for user/item profiling and rating prediction, and the recommendation model itself is more or less standard, not much different from other recommender systems, in this paper, most of the time we just list the algorithms without going into the detailed explanation of each algorithm. The explanation can be found in other survey papers.

This results in improved accuracy and user satisfaction [32, 143]. In the music domain, emotion-aware systems analyze lyrical emotions to suggest songs that align with users' moods. Using transfer learning techniques with models like BERT, these systems achieve enhanced performance in emotion detection and recommendation accuracy [144].

In the health domain, systems like emHealth [195] integrate emotion detection from self-rated depression data to recommend personalized psychological and physiological interventions. By considering user-specific emotional factors, these systems deliver actionable insights to improve mental health. Hotel and tourism recommendation systems incorporate emotion-driven statistical tools to rank and personalize services based on user reviews and emotional dimensions such as joy, anger, and trust [46, 188].

In dynamic domains such as the movie domain [39, 94], hybrid models that combine collaborative and content-based filtering with sentiment analysis of social media data can effectively capture emotional trends and user responses. For instance, in the work by Leung et al. [98], an advanced emotion classifier, the Tweets Affective Classifier (TAC), has been developed. TAC uses an asymmetric butterfly wing double-decker bidirectional LSTM-CNN Conv1D architecture to classify tweets into seven basic human emotions: happiness, sadness, disgust, fear, surprise, anger, and neutral. The output provides a probability distribution across these emotions, enabling precise emotional profiling of movie overviews. This emotional characterization facilitates personalized movie recommendations by aligning films with users' emotional preferences. Other recommenders utilize feature-sentiment pairing and emotion-tagging to align film recommendations with users' emotional states [39, 155].

In general, emotion-aware models demonstrate notable improvements over traditional methods. For instance, as reported in [155], integrating emotional features in movie recommendations has been shown to increase prediction accuracy by 173% for top recommendations. Additionally, emotion detection using probabilistic linguistic methods and deep learning architectures enables precise clustering and classification, facilitating robust and contextually relevant recommendations [143, 191]. These advancements highlight the critical role of emotional intelligence in enhancing user engagement and satisfaction across various application domains.

5.1 Algorithms

5.1.1 Algorithms for Emotion Detection. Emotion detection in text presents numerous challenges due to the inherent complexity and ambiguity of human languages. Factors such as sarcasm, irony, and cultural nuances often complicate the accurate identification of emotions. However, several algorithms have been developed to tackle these complexities by employing different strategies and methodologies to extract emotional content from text. Among these methods, sentiment analysis is one of the most widely used. Sentiment analysis aims to categorize the overall emotional tone of a text as positive, negative, or neutral, thereby offering a general understanding of the emotional content conveyed. By quantifying the sentiment polarity, this method helps determine whether a user expresses satisfaction, dissatisfaction, or indifference in a given context, which can then be used to enhance the personalization of recommendations [8, 10, 16, 35, 45, 79, 80, 94, 109, 133, 140, 142, 145, 155, 193].

Polarity classification is another method closely related to sentiment analysis, but it goes further by categorizing text into distinct emotional classes. Instead of simply determining whether a sentiment is positive or negative, polarity classification delves into the degree and nature of the sentiment, such as strong positivity, mild negativity, or neutrality, offering a more granular understanding of emotional expression [11, 32, 39]. It is also possible to leverage advanced algorithms like Bi-LSTM for sentiment analysis, enabling precise emotion classification and improved recommendation accuracy by integrating sentiment insights into collaborative filtering frameworks [80].

Other approaches are also utilized to refine the detection of emotions in textual data. POS tagging is one such approach, which involves analyzing the grammatical structure of sentences. By identifying and categorizing the words (e.g., nouns, verbs) in a sentence, POS tagging aids in the interpretation of emotional tone by providing

insight into how language is being used to express feelings or moods. For instance, certain combinations of adjectives and adverbs may intensify or mitigate the emotional impact of a statement, and POS tagging helps identify these patterns [39, 48].

Lexicon-based classification leverages predefined lexicons, which are sets of emotionally charged words, to classify text into various emotional categories. This method relies on matching words in the text with those in the lexicon to infer the emotional content, making it an effective tool for detecting emotions in specific contexts where lexicon coverage is robust [11, 38, 64, 101, 103, 110, 143, 163]. Semantic-based classification involves analyzing the structure and meaning of sentences, which can be used to detect the polarity or specific emotional tone of the text by breaking down and understanding its semantic components [193].

More sophisticated methods, such as Latent Dirichlet Allocation (LDA) topic modeling, are used to uncover hidden thematic structures in text that may be associated with specific emotions. By identifying and clustering topics within the text, LDA helps reveal the emotional themes present in large corpora of data. For example, topics associated with terms like “joy,” “happiness,” and “celebration” could be indicative of positive emotions, while topics centered around “grief,” “loss,” and “pain” might reveal negative emotional content [73, 104].

Bag-of-Words (BOW) models represent text by encoding the frequency of words within a document. This simplistic yet effective approach captures the presence of emotionally charged words and allows for the classification of text based on their distribution across different emotional categories. Although BOW models ignore the context and order of words, they can be useful in scenarios where the mere presence of specific emotional terms is sufficient for classification [60, 102, 199].

Finally, word embeddings offer a more nuanced method for capturing the emotional and semantic content of text. By mapping words to dense vector representations in a continuous space, word embeddings encode the relationships between words based on their context in a large corpus. This allows for the detection of subtle emotional nuances and similarities across different texts. For instance, words like “happy” and “joyful” may be close together in the embedding space, signaling a shared emotional connotation. Word embeddings thus enable more sophisticated emotion detection by capturing both semantic meaning and emotional similarity [48, 139, 163, 193]. Yang et al. [195] employed bidirectional LSTM-CNN Conv1D model to classify tweets into seven basic human emotions: happiness, sadness, disgust, fear, surprise, anger, and neutral.

In addition to these automatic methods, some manual methods have been used as well, such as self-assessment methods that offer a different approach by directly gathering user-reported emotions. Through the use of questionnaires or feedback mechanisms, users provide their own emotional responses to various stimuli or content. This self-reported data can then be integrated into the recommendation system, offering a personalized and highly accurate representation of the user’s emotional state, which is valuable for applications where subjective emotional experience is critical [65, 123, 195].

Another manual method employed in emotion detection is the analysis of review stars, particularly in systems where users rate products, movies, or other items. These ratings can be interpreted as indicators of emotional responses, with higher ratings often signifying positive emotions and lower ratings indicating negative sentiments. By correlating textual reviews with star ratings, recommendation systems can infer the underlying emotions associated with the user’s experience, even when the text itself may be ambiguous [11, 117].

5.1.2 Algorithms for Recommendation system. Table 3 presents a summary of the existing works using text-based data, classified by the algorithms used in their recommendation system.

5.2 Application Domains and Datasets

Textual datasets predominantly underlie recommender systems across various domains. Below, we explore their usage and the role of emotion in each domain. And Table 4 summarizes the commonly used datasets.

Table 3. Algorithms for Recommendation systems based on Text and Emotions

Categories	Algorithm	Publications
Traditional Models	Clustering	[39, 101, 103, 155, 190]
	SVM	[38, 45, 195]
	MF	[104, 117, 142]
	KG	[16, 64, 65, 145]
	CF	[10, 11, 98, 110, 123, 133]
	CBF	[60, 73, 98, 102, 109, 110, 140]
	Hybrid	[35, 94, 123, 142, 154]
Deep Learning Models	RNN	[38, 145]
	CNN	[8, 32, 48, 82, 193]
	MLP	[53]
	LSTM	[8, 193]
	Bi-LSTM	[80]
	GAN	[139]
	BERT	[70, 79, 144]
	Hybrid	[50, 111, 199]

- **Music:** When incorporating emotions into music recommendation systems, the goal is to identify the emotions conveyed by a piece of music and suggest it to users based on their emotional preferences or current mood. This can be achieved through various methods, for example, utilizing tags or labels assigned to music, measuring the similarity between songs based on their emotional features, examining user-generated text, such as social media posts or messages to infer the user's emotional state [163], or directly asking users about their emotions or having them select from predefined emotional categories [159].
- **Movie:** In the movie domain, the recommendation approach has largely relied on structured movie metadata to suggest films to viewers. This metadata typically includes information such as the director's name, genre, cast members, release year, Motion Picture Association (MPA) ratings, and keywords describing the movie's plot or themes. By analyzing these attributes, emotion is mostly taken into account based on the ratings of users' texts, MPA rating text, or user rating (1-5 stars). And then recommendation algorithms can identify similarities between movies and suggest titles that align with a user's preferences [32, 94, 156, 189, 204]. Systems also analyze user-generated content, such as user comments or reviews, to extract affective context, enhancing recommendation quality [131]. In addition to the metadata, other data can also be used. For example, Chien et al. [41] explored the use of movie plot summaries or synopses as an additional source of information for emotion identification. This approach recognizes that plot summaries can reveal nuanced details about a movie's narrative, characters, and tone that might not be captured by metadata alone.
- **Book:** In the book domain, the recommendation approach has primarily relied on structured book metadata to suggest titles to readers. This metadata typically includes information such as the author's name, genre, publication year, and ratings. By analyzing these attributes of text, they captured emotion and sentiment mostly based on keywords and ratings, and the employed recommendation algorithms can identify similarities between books and suggest titles that align with a user's emotion [80, 110, 142].

- **Social Media:** In the social media domain, the recommendation approach has often relied on user profile information, such as interests, demographics, and social connections, to suggest content, users, or groups. They mostly gather the textual content of posts, comments, and user interactions to capture the emotion of the users about specific things like a product, song, movie, etc. [39, 45, 104, 140, 160, 190, 193].
- **News:** In the news domain, the recommendation approach has involved analyzing article metadata like source, category, and keywords. Also, there is increasing emphasis on leveraging the textual content of news articles, including headlines, summaries, and full text, to do word embedding, or sentiment analysis [15, 17, 108, 135]. In addition, some use POI and questionnaires to capture user's emotions after reading news [123].
- **Product:** The product recommendation model has often relied on product attributes like category, brand, price, and user ratings to extract emotion information [10, 11, 102, 103, 117, 133], although some use tools such as questionnaires, rating scales, or feedback forms to gather subjective emotional data [102].
- **Tourism:** In the tourism domain, the recommendation approach has focused on destination attributes such as location, type (e.g., beach, city, historical), and popularity. They also use textual descriptions of destinations, reviews, and user-generated content to capture emotion and engage tourism recommendations [8, 38, 46, 70, 70, 73, 143, 143, 154, 188, 189, 191].
- **Health:** In health recommendation research, the traditional approach has involved analyzing patient medical records and demographic information of patient notes, symptom descriptions, and health-related online content to capture emotion from them. However, there is a growing trend toward using an application to log self-assessed emotion, feeling, and mood [34, 145, 195]. Nevertheless, Rumiantcev and Khriyenko [146] utilized music emotion, which is the annotation based on their metadata such as artist name, track name, release date, genre, and lyrics to reduce anxiety.
- **Education:** In the education domain, the recommendation approach has relied on student profiles, learning styles, and performance data, course descriptions, learning materials, and student feedback to provide more adaptive and personalized educational recommendations. They mostly capture students' emotion from their reviews or posts on social media [36, 50, 111].

6 Emotion-aware Image-Based Recommender Systems

EIRs have evolved through diverse methodologies and algorithms. This section reviews key contributions from recent studies, highlighting advancements in emotion-driven recommendation systems and the algorithmic approaches that enable real-time emotion detection and personalized recommendations.

6.1 Facial Emotion Detection in Image-Based Systems

Ali et al. [5] and Angkasa et al. [9] focused on emotion detection using features extracted from facial images, achieving high accuracy on the FER2013 dataset. Ali et al. employed PCA and LDA, classical feature extraction and dimensionality reduction techniques that identify the most discriminative features in facial image data by projecting high-dimensional pixel information into lower-dimensional feature spaces that separate emotions effectively, for emotion-based music recommendations. These methods learn mathematical transformations that maximize the variance between different emotions while minimizing variance within the same emotion class, enabling efficient classification with interpretable results. In contrast, Angkasa et al. implemented a CNN + XGBoost, which combines deep learning with classical machine learning. The CNN component automatically extracts hierarchical visual features from facial images through multiple convolutional layers, learning progressively abstract features from low-level patterns (edges, textures) through mid-level features (facial parts like eyes and mouth) to high-level semantic features (emotional expressions). This hierarchical feature learning mimics human perception and has proven highly effective for facial expression recognition. The extracted features are

Table 4. Most Commonly Used Datasets in Text-Based Emotion-Aware Recommender Systems (ETRSs)

Domain	Dataset	Detailed Description & Characteristics
E-commerce	Amazon Product Reviews	142.8M reviews (May 1996–July 2014); 1–5 star ratings with review text; emotion signals: sentiment polarity (positive/negative/neutral) + aspect-based opinions; available at https://cseweb.ucsd.edu/~jmcauley/datasets.html
	Yelp Reviews	8.6M reviews, 200K businesses; 1–5 star ratings; emotion labels: positive/negative/neutral sentiment + temporal dynamics; available at https://www.yelp.com/dataset
	Epinions	~1.56M product reviews (electronics, books, movies); 1–5 star ratings with trust network data; emotion signals: sentiment polarity + trust/distrust indicators; archived versions available
Social Media	Twitter Emotion Dataset	40K tweets with 6 basic emotions (joy, sadness, anger, fear, surprise, love) + neutral; collected via Twitter API; available at https://github.com/dair-ai/emotion_dataset
	Reddit Posts	100K–1M posts per study; subreddit-specific emotional contexts; emotion labels vary by community; accessible via Reddit API or Pushshift
	Facebook Reactions	User reactions (Like, Love, Haha, Wow, Sad, Angry) mapped to 6 discrete emotions + engagement metrics; proprietary, available via research partnerships
	myPersonality	22,500 users, 19M status updates; Big Five personality traits + emotional expression patterns; available to approved researchers at http://mypersonality.org/wiki/
Entertainment (Movies)	MovieLens 25M	25M ratings from 162K users on 62K movies; 1–5 star scale; emotion inferred from ratings, tags (e.g., “funny,” “scary”), reviews; available at https://grouplens.org/datasets/movielens/25m/
	IMDb Reviews	50K highly polar movie reviews (25K positive, 25K negative); binary sentiment labels; emotion signals: polarity + intensity; available at https://ai.stanford.edu/~amaas/data/sentiment/
	LDOS-CoMoDa	2,296 ratings with contextual metadata; unique feature: 22 distinct user emotions (e.g., dominant, afraid, satisfied, lonely); includes time, location, social context; available at http://ldos.si-lab.org/
Music	Last.fm-1K	992 users, 176K artist listening events; implicit emotion via listening patterns (skips, repeats, duration); tags like “happy,” “chill,” “intense”; available at http://ocelma.net/MusicRecommendationDataset/
	Music4All	109K songs with 4-point affective scale; emotions rated on valence-arousal dimensions; includes audio features (tempo, key, energy); available at https://github.com/Music4All/music4all
	MER-GEMS	364 music excerpts with Geneva Emotional Music Scale (GEMS) annotations; 9 dimensions: wonder, transcendence, tenderness, nostalgia, peacefulness, power, joyful activation, tension, sadness; available upon request
	DEAM	1,744 songs with continuous valence-arousal annotations; dynamic (per-second) emotion ratings + audio features; available at https://cvml.unige.ch/databases/DEAM/
News	News Category (HuffPost)	200K+ headlines (2012–2018); 41 categories; emotion signals: sentiment + topic-emotion associations; available at https://www.kaggle.com/datasets/rmisra/news-category-dataset
	MIND (Microsoft)	160K English news articles, 1M user impressions; implicit emotion from clicks + article sentiment; includes demographics; available at https://msnews.github.io/
Tourism	TripAdvisor	10K–1M reviews per study; 1–5 star ratings with geo-spatial metadata; 8 emotional dimensions (joy, anger, disgust, fear, sadness, surprise, trust, anticipation); web-scraped datasets on Kaggle
	Booking.com	Multi-lingual reviews (50+ languages); 1–10 rating scale; emotion labels: positive/negative + aspect-based emotions; research access via partnerships
Health	emHealth Dataset	Self-rated depression scores (PHQ-9) + health diary text; emotion labels: depression severity (none, mild, moderate, severe); N=1,200 participants; available upon request
	Mental Health Reddit	187K posts from r/anxiety, r/depression, r/SuicideWatch; binary stress labels + emotion-laden text; available at https://ir.cs.georgetown.edu/resources/dreaddit.html
Education	MOOC Learner Comments	~50K student forum posts from Coursera/edX; emotions: confusion, frustration, boredom, engagement; available via educational data mining repositories
	Course Reviews	100K+ reviews from Rate My Professors; 1–5 ratings + qualitative feedback; emotions: satisfaction, frustration, enthusiasm; web-scraped datasets on GitHub
Product (General)	Amazon Electronics	7.8M reviews; emotion signals: sentiment polarity + product feature-emotion associations; available at https://cseweb.ucsd.edu/~jmcauley/datasets.html
Books	Goodreads	15M reviews across 2M books; 1–5 star ratings + review text; emotion signals: sentiment + genre-emotion correlations; accessible via Goodreads API
	Amazon Books	22.5M reviews (1996–2018); emotion: rating-based sentiment + NRC Emotion Lexicon extraction; available at https://cseweb.ucsd.edu/~jmcauley/datasets.html

then fed into XGBoost, an ensemble machine learning method that combines predictions from multiple decision trees with adaptive weighting, prioritizing difficult-to-classify samples. This hybrid approach leverages CNN's strength in automatic feature discovery with XGBoost's robustness and interpretability, addressing the incomplete cold-start problem with an impressive accuracy of 80.99%. Similarly, Mishra et al. [122] integrated CNNs for real-time emotion detection and personalized music playlist generation. CNNs are particularly well-suited for real-time applications because they can efficiently process images through local connectivity and weight sharing, convolutional filters of small size (e.g., 3×3 or 5×5 pixels) are applied across the entire image, dramatically reducing the number of parameters compared to fully connected networks. This computational efficiency enables deployment on mobile devices and real-time processing of video frames, making CNNs the standard approach for facial emotion recognition in practical systems. The learned convolutional filters capture emotion-relevant patterns (facial muscle configurations, expressions) automatically from training data without manual feature engineering.

Pauly et al. [137] proposed a novel product recommendation system that combines multiple algorithmic components. They employed the Eigenfaces algorithm, a classical approach in facial recognition that uses PCA to learn the principal components of facial images, the "faces" that represent the most significant variations across the face database. Eigenfaces enables efficient face detection and recognition by comparing test faces to these principal components, providing computational efficiency crucial for real-time systems. Complementing this, they used the Fisherfaces method for emotion detection, an extension of LDA specifically designed for facial feature analysis that maximizes between-class variance (differences between emotions) while minimizing within-class variance (variations within the same emotion), yielding more discriminative emotion classification than Eigenfaces alone. Their approach integrates gender identification to enhance targeted product recommendations and implements a feedback loop based on user emotions, dynamically adjusting suggestions to ensure user satisfaction, a prototypical example of feedback-driven emotion-aware recommendation systems. Several studies explored real-time systems leveraging facial emotion detection for music recommendations using specialized computer vision algorithms. Iyer et al. [71] developed EmoPlayer, an Android-based music recommender using the Viola-Jones algorithm for face detection and the Fisherfaces classifier for emotion recognition. The Viola-Jones algorithm, a cascade classifier trained with AdaBoost, achieves rapid real-time face detection by efficiently evaluating Haar features (simple rectangular filters that compute differences between light and dark image regions) in a cascade hierarchy, rejecting non-face regions early to dramatically reduce computation. This makes Viola-Jones the standard for mobile and embedded face detection despite newer deep learning methods. Once faces are detected, Fisherfaces provides robust emotion classification. Notably, EmoPlayer implements an adaptive recommendation strategy: the system detects the user's current emotional state and transitions them toward positive emotional states through personalized music selection, a practical instantiation of emotion-aware recommendation logic.

Singh et al. [166] introduced a more modern system integrating CNNs with Spotify API for emotion-based playlist suggestions via facial expressions. By leveraging CNNs' superior feature learning capacity compared to hand-crafted features (Haar cascades, Eigenfaces), the system achieves more nuanced emotion recognition and can recommend music with greater personalization. The integration with Spotify's music database and API demonstrates the practical deployment of CNN-based emotion detection in real-world music streaming services.

Vayadande et al. [183] used the Haar Cascade Classifier, a cascade of classifiers trained on Haar features, similar to Viola-Jones but often implemented through OpenCV libraries, for face detection and SVM for emotion classification. SVMs are classical supervised learning algorithms that find an optimal separating hyperplane in feature space to maximize the margin between different emotion classes. When combined with hand-crafted facial features (e.g., Histogram of Oriented Gradients, Geometric features), SVMs provide robust and interpretable emotion classification. Their approach achieved 84.82% accuracy, demonstrating that classical machine learning can be highly effective when combined with appropriate feature engineering.

Shalini et al. [162] and Bhutada et al. [24] emphasized real-time implementation using specialized facial recognition algorithms. Shalini et al. leveraged OpenCV for facial emotion detection and Pygame for real-time music playback, demonstrating a complete pipeline from emotion detection to audio output. Bhutada et al. processed facial features to match emotions such as happiness or sadness, requiring explicit feature extraction or implicit learning through neural networks, and tailored music playlists accordingly. These studies highlight the engineering considerations in deploying EIRS systems: computational efficiency (real-time processing), system integration (linking emotion detection to audio), and user experience (dynamic adaptation to emotional state).

Beyond facial expression, image-based emotion understanding extends to aesthetic and style domains. Becattini et al. [23] employed **Kobayashi's Color Image Scale**, a psychological framework linking specific color combinations to emotions and styles (e.g., "casual," "elegant," "romantic"), to classify styles based on emotional associations of color patterns. For computational classification, the system uses ResNet18, a deep convolutional neural network architecture with skip connections (residual blocks) that enable training of very deep networks. Residual blocks allow gradients to flow directly through skip connections, solving the vanishing gradient problem that plagued earlier deep networks. Each residual block learns an incremental transformation (residual) rather than the full transformation, enabling the network to be 10-20 times deeper than networks without skip connections while remaining trainable. ResNet's depth enables capture of hierarchical visual features essential for nuanced style understanding: low layers learn color patterns and textures; mid layers learn garment shapes and arrangements; deep layers learn semantic style attributes. The ResNet18 model learns from labeled examples of outfit combinations, achieving a notable accuracy of 78.56% in style classification, effectively mapping color patterns to associated emotions and styles. This success demonstrates that deep CNNs can learn implicit emotional associations in visual aesthetics without explicit emotion labels, learning style-emotion mappings from data rather than from manually defined rules. Following the initial style classification, Becattini et al. further developed **GR-MANN (Memory-Augmented Neural Network)**, which stores non-redundant garment samples to suggest compatible items (e.g., matching tops and bottoms). The memory augmentation mechanism maintains a set of representative garment samples, enabling efficient similarity-based retrieval while reducing storage redundancy. GR-MANN enhances recommendations by combining outputs from style and event classifiers, tailoring suggestions to users' preferences and occasions. This showcases the diverse strategies in EIRSs, demonstrating how systems can adapt to users' emotions and preferences for highly personalized fashion recommendations.

The studies collectively demonstrate the evolution of emotion detection algorithms in EIRSs, from classical feature extraction methods (PCA, LDA, SVM) through modern deep learning approaches (CNNs, ResNets) to emerging architectures (memory-augmented networks). Classical methods (PCA, LDA, Fisherfaces, SVM) offer interpretability, computational efficiency, and strong performance with feature engineering, making them ideal for resource-constrained or interpretability-critical applications. Modern CNNs and ResNets enable end-to-end learning from raw images, achieving superior accuracy and handling complex visual patterns without manual feature design, at the cost of increased computational requirements and reduced interpretability. Specialized architectures like ResNet and memory-augmented networks demonstrate that different emotion-aware tasks (facial expression, style classification, recommendation generation) benefit from tailored algorithmic approaches. The progression of systems studied, from PCA/LDA (1990s-2000s) through CNNs (2010s) to residual networks and memory-augmented approaches (2020s), reflects the broader machine learning revolution: moving from hand-crafted features to learned representations, from shallow classifiers to deep networks, and from task-specific systems to modular, adaptable architectures. This evolution enables EIRSs to deliver increasingly personalized, real-time, emotion-driven recommendations across diverse domains (music, products, fashion), with improved accuracy, efficiency, and user experience.

Table 5. Dataset utilized in EIRSs

Publication	Dataset	Dataset Explanation
[5, 9, 122, 166]	FER 2013	Images of various facial expressions labeled with corresponding emotions
[71]	CK+ Database	Images of different emotions used for facial expression recognition
[88]	IAPS	Images with normative ratings of valence (pleasantness) and arousal
[162]	author gathered dataset	facial image dataset
[24]	author gathered dataset	facial image dataset
[183]	Author gathered dataset	facial image dataset while watching music video
[23]	Fashion4Events dataset	clothing event image dataset for emotion-based outfit recommendations
[137]	Author gathered dataset	Facial image dataset with emotion (happy/sad) for product recommendation
[92]	The SFEW Database	Photos of seven facial expression classes under challenging conditions.

6.2 Application Domains and Datasets

EIRSs analyze visual content, such as facial expressions, visual aesthetics, and the mood conveyed by images, to align recommendations with users' emotional preferences or current moods. Key application domains and datasets utilized in this field are as follows:

- **Music:** EIRSs are primarily used to analyze people's emotions by interpreting their facial expressions recorded while they listen to the music [5, 9, 24, 71, 122, 162, 166, 183].
- **Fashion:** Emotion-aware systems recommend clothing styles that align with users' emotional preferences, such as soothing designs for a calm mood or vibrant styles for an energetic feel. These systems enhance the shopping experience by personalizing fashion choices based on mood and aesthetic appeal [23].
- **E-commerce:** Emotion-aware systems enhance the online shopping experience by recommending products with strong visual appeal and emotional resonance. By analyzing emotional reactions to product images, these systems ensure a more engaging and satisfying shopping journey [137].
- **Social Media:** On image-sharing platforms, emotion-aware systems recommend images based on user preferences and emotional states, fostering more relevant and emotionally resonant content discovery [88].

Table 5 summarizes the commonly used datasets for developing and evaluating EIRSs.

7 Emotion-aware Audio-Based Recommender System

Emotion-aware audio-based recommender systems leverage the unique ability of audio (mostly music) to influence and regulate human emotions. These systems analyze key musical attributes such as tempo, rhythm, harmony, and timbre to identify the emotional content of audio. By collecting user data, including mood and preferences, these systems generate personalized recommendations that align with users' emotional needs.

Techniques such as Music2Vec [30] convert raw audio into embeddings that capture a song's emotional content, making it easier for models to process. LSTM networks are then applied to these embeddings to predict users' emotional states over time, leveraging the sequential nature of music data. For instance, Liu et al. [101] developed an LSTM-based framework that recommends slightly positive songs to improve users' mental states. Shepstone et al. [164] introduced a novel TV recommendation system that uses audio-derived user emotions to enhance personalization. The main innovation is the "affective offset" framework, which assesses the gap between a user's perceived emotional state and the emotional characteristics of the recommended content. Moscato et al. [126] integrated psychological insights such as the Big Five personality traits into a content-based framework, significantly enhancing the personalization and emotional relevance of recommendations. Finally, Strauss et al. [172] offered a comprehensive dataset of 364 music excerpts annotated with the Geneva Emotional Music Scale, facilitating accurate emotion-to-music mapping and furthering research in music emotion recognition (MER) and personalized music recommender systems to improve mental health.

7.1 Algorithms

EARSs employ diverse algorithms for emotion detection and music recommendation. Some representative ones are reviewed in this section.

7.1.1 Algorithms for Emotion Detection. Emotion-aware audio-based recommender systems use diverse methods to identify and model user emotions. Music2Vec [30] extracts emotional embeddings directly from audio data, while Mel-Spectrograms [126] analyze audio features visually to infer emotional content from songs. Advanced sequential models like LSTM networks effectively capture long-term dependencies in music data, enabling the prediction of evolving user emotional states over time [101]. Interaction-based systems, such as those developed by Yousefian et al. [197], analyze patterns in user interactions like mouse and keyboard usage to infer emotional states, bypassing explicit detection methods and reducing errors. Shepstone et al. [164] used I-Vector Modeling which enhanced emotion detection by extracting acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs) and classifying them into emotional categories using Probabilistic Linear Discriminant Analysis (PLDA).

7.1.2 Algorithms for Recommendation systems based on Audio and Emotions. Recommender systems leverage algorithms to match content with user preferences and emotional states. Hybrid models that combine Collaborative Filtering (CF) and Content-Based Filtering (CBF) integrate psychological insights to consider user preferences, emotional states, and personality traits, offering highly tailored recommendations [126]. Sequential models like LSTMs are adapted for recommendation tasks, using temporal emotional patterns to enhance suggestions [101]. Additionally, Shepstone et al. [164] introduces a unique Circumplex Model of Affect, which maps content and user moods into a Valence-Arousal (VA) space. Recommendations are refined interactively through critique-based navigation, allowing users to adjust recommendations by specifying preferences for valence (pleasantness) and arousal (intensity). This paper also employs an Affective Offset Adjustment algorithm, which calculates the offset between initial and final user preferences in the VA space, to improve future recommendations.

7.2 Application Domains and Datasets

EARSs have been applied across various domains by leveraging datasets that capture the emotional characteristics of music. Table 6 summarizes key datasets used for developing and evaluating EARSs. The primary application domains are outlined below:

Table 6. Summary of datasets with their explanations and citations.

Citation	Dataset	Explanation
[7, 101, 172]	DEAM Dataset	Includes songs labeled with emotional valence and arousal, enabling mapping of musical features to emotional states.
[7]	Million Song Dataset	A large collection of songs with features such as tempo, rhythm, and harmony, often used to improve music recommendation accuracy.
[172]	EMMA	Includes 364 music excerpts annotated with emotional ratings based on GEMS.
[101]	NetEase Cloud Music Dataset	Used alongside the GTZAN dataset, capturing user interactions and preferences for exploring genre-specific emotional responses.
[126]	Deezer Dataset	Contains 18,645 songs with metadata such as valence and arousal ratings. Used for mood detection and recommendation tasks.

- **Music Streaming Services:** These systems recommend songs based on users' moods or emotional states. For example, users feeling sad might receive calming or uplifting recommendations. Such systems often utilize embeddings from audio features and models trained on DEAM and the Million Song Dataset [7, 112].
- **Mental Health Therapy:** These systems are used in clinical settings to recommend music that stabilizes or enhances mood. For instance, the Emotion-to-Music Mapping Atlas (EMMA) facilitates precise music recommendations tailored to improve mental states [172].
- **Media Recommendation:** These systems are designed for TV (or other media types) content recommendations where user emotions and moods affect viewing preferences [164].

8 Emotion-aware Video-based Recommender Systems

EVRs represent a novel approach to enhancing user experiences in video content recommendations by integrating emotional analysis. Unlike traditional systems that focus solely on user preferences derived from historical interactions, EVRs delve deeper by analyzing emotional states through a variety of signals. These signals include physical signals like facial expressions, body language, and voice tone. By interpreting these emotional cues, EVRs aim to provide tailored and contextually relevant recommendations that resonate with a viewer's current emotional state. Physical signals, often captured through webcams, can be processed using advanced machine learning algorithms that analyze variations in expressions, postures, and vocal inflections. These algorithms are designed to recognize patterns associated with specific emotions, allowing EVRs to discern subtle shifts in a viewer's mood. For instance, a smile may indicate enjoyment or satisfaction, while crossed arms could suggest defensiveness or discomfort.

Techniques such as facial expression analysis and body gestures enable EVRs to classify emotions with high precision in valence-arousal-dominance spaces [169]. Advanced hybrid models, like ResNet50 combined with CNN-LSTM frameworks, further enhance temporal and spatial emotion recognition by leveraging video data [176]. Bandyopadhyay et al. [21] proposed a real-time emotion detection system that uses facial expressions captured through a webcam. By employing a CNN, their system detected emotions such as happiness, anger, sadness, and surprise from facial images. It offered real-time product recommendations during online shopping based on the user's emotional state, enabling highly interactive and emotionally aware recommendations without relying on historical data or purchase records. Choi et al. [42] similarly proposed a facial expression-based collaborative filtering system that addresses the new user problem by creating dynamic user profiles from real-time facial expression data instead of relying on historical user ratings or interactions.

8.1 Algorithms

8.1.1 Algorithms for emotion detection. Emotion detection in EVR leverages physical signals captured in videos to interpret users' emotions in real-time while interacting with the items. Here are the key algorithms used for emotion detection in EVR:

- **SVM:** SVM is widely used in EVR to classify users' emotional states based on input features such as facial expressions and voice. For instance, SVM models can classify emotions into predefined categories such as happy, sad, or neutral by training on labeled datasets. Arapakis et al. [13] trained a hierarchical SVM model to predict video relevance based on real-time facial expressions.
- **AdaBoost:** AdaBoost algorithms are employed to improve the accuracy of emotion detection models by combining weak classifiers into stronger classifiers. Zhao et al. [201] enhanced AdaBoost to recognize facial expressions for affective video recommendations.
- **Matrix Factorization Algorithms:** Matrix Factorization algorithms create decision matrices combining users' emotional states and other features for emotion detection. These algorithms are particularly useful when

Table 7. Emotion-aware Video-based Recommendation System Categorization based on Algorithm

Categories	Algorithm	Publications
Traditional Models	Regression	[52, 167]
	Clustering	[129]
	SVM	[68, 72, 118, 130, 167, 170]
	AdaBoost	[165]
	KNN	[137]
	KG	[28, 141, 186]
	CF	[40, 42, 169, 179]
	CBF	[49, 180]
	Hybrid	[113, 184]
	MF	[75, 76]
Deep Learning Models	RL	[96, 182, 200]
	CNN	[9, 21, 120]
	LSTM	[185]
	MLP	[19, 27]
	GAN	[120]
	Hybrid	[56, 181, 205]
Generative Models		[62, 202]

handling complex datasets that require simultaneous evaluation of multiple user parameters. Kaklauskas et al. [75, 76] applied matrix factorization algorithms to analyze data for video recommendations.

- CNN: CNNs are frequently applied for feature extraction in emotion recognition, particularly for analyzing facial expressions [20]. CNN models can automatically learn to identify patterns in visual data associated with emotional states.
- Reinforcement learning: RL models in EVRS adaptively learn strategies based on users' real-time state changes in response to video content. These models continuously learn from user behavior and adjust video recommendations to maximize engagement by considering emotional states. Tripathi et al. [181, 182] employed RL to monitor viewers' cognitive preferences and behaviors in real-time.

8.1.2 Algorithms for recommendation system. We have categorized the algorithms used in EVRSs in Table 7.

8.2 Application Domains and Datasets

EVRSs have been employed across various domains to enhance user experiences by capturing and interpreting emotions through different methods. Below, we explain how EVRS have been utilized in several domains, detailing the techniques used to capture and interpret user emotions for personalized recommendations.

- Music: In the music domain, EVRS capture emotions through features like facial expressions and sentiment analysis. For instance, systems can analyze facial cues during music playback to infer emotional states, recommending upbeat tracks for positive emotions or calming music for stress relief. Emotion recognition is enhanced by integrating user feedback and sentiment analysis of playlists or tracks [6, 9, 20, 127, 162, 186].
- Movie: Movie recommendations in EVRS utilize a combination of facial expression analysis and textual feedback such as reviews or ratings. For example, systems can detect emotional cues through real-time

Table 8. EVRS Dataset Categorization on Different Domains

Dataset	Explanation
Moviepilot Mood Track	4.5M ratings by 105K users on 25K movies, with contextual information.
The Five Emotional Reactions	Webcam recordings of expressions: happiness, sadness, anger, fear, surprise.
The Clicker and Emotional Reaction	Webcam recordings and survey data of subjects watching videos.
LDOS-PerAff-1 Corpus	Affective responses to video clips in VAD space, with personality data.
Mechanical Turk Setup	Affective annotations of video clips for boredom evaluation.
Cohn-Kanada Database	Photos of people expressing six basic emotions, from neutral to intense.
The AFEW Database	Short video clips of facial expressions in real-world scenarios.
The Hollywood Movie Video	Movie clips annotated with emotions.
The Tellyads and YouTube Video	Videos from various genres (TV shows, movies, news).
The Affective Property Movie	Videos with arousal and valence ratings.
Nvidia 3D Vision Database	Stereoscopic video sequences.
The Movie Profile Database	Movie profiles with various content-related attributes.
Algebra Video Field Test Database	Data collected from students and teachers watching educational videos.
The User Action Session Database	User preference features extracted from videos.
The Format Video Database	Videos from various websites.
The Footwear Advertising Videos	User facial features and ratings while watching footwear advertising videos.
The NEAR Database	Collection of databases, including video clips and text data.
LIRIS-ACCEDE	Feature films and short films with emotional labels.
PM-SZU	Video clips extracted from films for video emotion-aware analysis.
imdb.com Database	Movie reviews from imdb.com.
YouTube Video Clips	Videos with corresponding transcripts.
IMDB Movie Scenes	Duration data of users watching movie scenes on IMDB.

facial expression tracking via webcams, which are then classified into categories such as happiness, sadness, and surprise. These insights help recommend movies that align with the viewer's preferences and emotional states [28, 75, 129, 205].

- **Education:** In educational settings, EVRS can adapt video recommendations based on students' emotional responses to learning content. Emotions like anxiety, boredom, or engagement can be detected through facial expressions. For instance, if a student shows signs of frustration or stress while watching a difficult lesson, the system might recommend simpler or more motivational videos to reduce anxiety and enhance focus [96].
- **Video:** Online video platforms can leverage EVRSs for real-time, emotion-driven recommendations, particularly for short advertising videos. By capturing viewers' emotional responses, these systems enhance user engagement and deliver more personalized viewing experiences [42].
- **Product:** In the product domain, EVRS can recommend products by analyzing emotional reactions to product videos or advertisements. For example, facial expression analysis can detect user emotions like excitement or satisfaction during a product demonstration, guiding recommendations for similar products or brands. Sentiment analysis of comments and reviews further refines recommendations by understanding user preferences and emotional connections [21, 72, 124].
- **Book:** Although books are text-based, EVRS can recommend book-related videos such as trailers, interviews, and reviews by analyzing emotional reactions. Techniques like facial expression recognition and sentiment analysis of user comments help determine emotional alignment with specific genres or themes. For instance, a user showing excitement during a book trailer may be recommended other books or genres that evoke similar responses [114].

- **Tourism:** EVRS in the tourism domain rely on detecting emotional reactions to travel-related videos, such as destination overviews or activity showcases. Methods like gaze tracking and facial recognition allow the system to interpret the emotional impact of various scenes. For instance, a video about hiking in the mountains might trigger excitement or relaxation, leading the system to recommend similar adventurous or calming activities and destinations [151].
- **Fashion:** In fashion, EVRS capture emotions through real-time analysis of facial expressions during fashion shows or product videos. For example, if a user exhibits positive emotions like confidence or joy after watching a video featuring a certain clothing style, the system can recommend similar fashion items that align with the user's emotional response. These insights help users discover fashion choices that enhance their mood and self-expression [107].

There are various datasets that are essential for research in EVRSs in different domains. Data collection methods vary, ranging from capturing changes in viewers' facial expressions via webcams to analyzing sentiment through user comments. These datasets and their details are listed in Table 8.

9 Emotion-Aware Physiological Recommender Systems

EPRSs have seen significant advancements, leveraging diverse methodologies and signal-processing techniques to enable emotion-driven personalization. Physiological responses, triggered involuntarily by the central and autonomic nervous systems, offer reliable data for emotion analysis. This section reviews key contributions from recent studies, highlighting innovations in physiological data utilization for personalized recommendations.

Santamaria et al. [153] and Kwon et al. [95] utilized EEG and ECG signals for emotion recognition, demonstrating accuracy improvements over traditional methods. Santamaria et al. employed deep CNNs to analyze ECG and GSR data for healthcare applications, while Kwon et al. proposed a fusion-based 2D CNN architecture to process EEG spectrograms and GSR features for emotion classification.

Real-time emotion detection systems have gained prominence, with innovations such as EmoStream by Tripathi et al. [181], which uses reinforcement learning to adapt content recommendations based on physiological signals dynamically. Shu et al. [165] developed a stress-aware movie recommender system by monitoring continuous heart rate and GSR, enhancing adaptability to users' emotional states. Dabas et al. [47] explored EEG-based emotion-aligned tourist video recommendations, while Mishra et al. [121] proposed stress-reducing content recommenders using HRV analysis.

Guo et al. [63] introduced a hybrid framework that integrates prepurchase and postpurchase ratings to improve preference modeling. Alhagry et al. [4] combined EEG and voice data with deep learning to generate real-time emotion-driven music playlists. Kalaganis et al. [77] proposed Musical NeuroPicks, a BCI system leveraging EEG biomarkers to quantify user responses to music, offering high accuracy for real-time music recommendations.

Yang et al. [194] addressed the non-stationary nature of EEG signals by introducing a multi-column CNN architecture that aggregates temporal EEG data for emotion classification. Zhang et al. [198] developed EEG-SVRec, a novel dataset combining EEG signals and multidimensional affective engagement scores for short video recommendations, setting a benchmark for affective video systems. Santamaria et al. [151] utilized ECG-based emotion recognition in low-cost wearable devices to provide tourist experience recommendations. In another study, Santamaria et al. [153] highlighted the effectiveness of deep learning in healthcare applications.

9.1 Algorithms

9.1.1 *Algorithms for emotion detection.*

- **Traditional Machine Learning Models:** Santamaria et al. [151] utilized traditional machine learning classifiers, including Decision Trees and SVM, to analyze GSR and ECG features for emotion recognition.

- **Frequency Band Analysis:** Kalaganis et al. [77] analyzed EEG signals using cross-frequency coupling and activation asymmetry indices to detect emotional states. Zhang et al. [198] employed Differential Entropy (DE) to extract features from five EEG frequency bands (delta, theta, alpha, beta, and gamma), improving emotion classification accuracy.
- **CNN and LSTM Approaches:** Santamaria et al. [153] utilized deep CNN models to analyze ECG and GSR data for emotion detection. Yang et al. [194] introduced a multi-column CNN architecture to capture temporal features in EEG data, achieving robust emotion classification. Alhagry et al. [4] employed LSTM layers to model temporal dependencies in EEG signals, enabling effective classification of valence and arousal levels. Similarly, Mishra et al. [121] explored pre-trained CNNs like GoogLeNet and AlexNet for emotion detection, demonstrating the power of transfer learning in this domain.

9.1.2 Algorithms for recommendation system. Recommender systems in emotion-aware applications employ a variety of algorithms tailored to integrate physiological signals and emotional data for personalized recommendations. Traditional collaborative and content-based filtering techniques have been adapted to incorporate emotion data. For example, Mishra et al. [14] combined collaborative and content-based filtering with emotion data derived from physiological signals. Reinforcement learning has been used to dynamically enhance content-based recommendation systems by incorporating user feedback, as demonstrated by Alhagry et al. [18].

Advanced machine learning models have also been benchmarked for emotion-based recommendations. Zhang et al. [198] evaluated frameworks like Factorization Machines (FM), DeepFM, and Wide&Deep, with and without EEG signals, highlighting the impact of physiological data on recommendation performance. Other techniques such as Extreme Learning Machines are used for music recommendation [77], which leverage EEG data to predict user preferences with high accuracy.

9.2 Application Domains and Datasets

EPRSs span various application domains, leveraging diverse datasets for training and evaluation. Below is an overview of key domains. The datasets utilized in recent studies are listed in Table 9.

- **E-Commerce:** Guo et al. [63] developed a hybrid recommender integrating prepurchase and postpurchase ratings using a proprietary dataset to improve customer preference modeling.
- **Music Recommendation:** Alhagry et al. [4] and Kalaganis et al. [77] used EEG data from DEAP and other sessions to predict music preferences.
- **Video Recommendation:** Zhang et al. [198] used EEG-SVRec, a real-world short video dataset with EEG and ECG data, while Mishra et al. [121] utilized DEAP for emotion-driven video tagging.
- **Healthcare:** Santamaria et al. [153] employed AMIGOS and DEAP datasets to classify emotions for stress-aware systems.
- **Tourism:** Santamaria et al. [151] used proprietary HR data from wearable devices to recommend tourist experiences based on emotional states.

Table 9. Application Domains and Datasets for Emotion-Aware Physiological Recommender Systems

Domain	Dataset	References
E-Commerce	Proprietary prepurchase and postpurchase data	[63]
Music Recommendation	DEAP, EEG from music sessions	[4, 77]
Video Recommendation	EEG-SVRec, DEAP	[121, 198]
Healthcare	AMIGOS, DEAP	[95, 153]
Tourism	Proprietary HR data from wearables	[151]

10 Emotion-Aware Multimodal Recommender Systems

EMRSs integrate data from diverse modalities, such as facial expressions, speech, text, and physiological signals, to deliver highly personalized and contextually relevant recommendations. By analyzing and interpreting emotional states through advanced datasets and algorithms, these systems significantly enhance recommendation accuracy and user experience. For instance, Arapakis et al. [12] developed a video recommender system that uses real-time emotional feedback from facial expression analysis and text-based information, improving accuracy from 78% to 89%, using a hierarchical SVM model. Zhao et al. [203] introduced early and late fusion models to combine audio and text embeddings, enabling the system to process nuanced emotional cues from speech and sentiment for more robust emotion recognition. Diaz et al. [51] introduced a noninvasive affective video recommendation approach that uses standard webcams to collect emotional data (facial expressions and heart rate) without requiring wearable devices.

Yerkin et al. [196] proposed a group movie recommender system leveraging text, audio, and image modalities, employing a fuzzy inference system to cater to collective emotional preferences. Similarly, Kim et al. [89] combined SVMs with collaborative filtering to detect speech-based and text-based emotions and match multimedia content to emotional states like happiness or sadness. The ARTIST system by Kaklauskas et al. [74] combined EEG signals with facial expression data for personalized learning material recommendations, achieving state-of-the-art accuracy in recognizing learner emotions. Bakariya et al. [20] developed an affective video recommender system leveraging facial expressions and pulse signals to predict user preferences based on affective data during video viewing. Tkalcic et al. [177] enhanced image recommendations by integrating affective metadata from facial expressions and physiological signals using the IAPS dataset, improving precision. Krupa et al. [93] leveraged CNNs for real-time emotion recognition by analyzing facial expressions and chatbot interactions, aligning recommendations with detected emotions.

In the domain of music recommendation, Rumiantcev et al. [146] designed a system for emotional and physical well-being by leveraging AI and traditional music therapy approaches. Hossain et al. [69] proposed an audio-visual emotion recognition system using CNNs for speech and video signals, achieving high accuracy through ELM-based fusion on large datasets. Abdul et al. [1] introduced a music recommender system using weighted feature extraction (WFE) and DCNNs to identify latent emotional features, significantly enhancing music personalization. Similarly, Djordjević Čegar et al. [132] applied psychophysiological signals and text with a multilayer perceptron neural network to estimate arousal and valence in stereoscopic 3D video content.

Aivazoglou et al. [2] focused on fine-grained social network recommendations by incorporating sentiment analysis and visual content to deliver emotionally aligned multimedia suggestions. Hossain et al. [69] further enhanced emotion recognition and recommendation accuracy by fusing audio and video data using Mel-spectrograms, 3D CNNs, ELMs, and SVMs. Collectively, these approaches demonstrate the transformative potential of multimodal systems in delivering emotionally aware, personalized, and user-centered recommendations.

10.1 Algorithms

10.1.1 Algorithms for emotion detection. Recent studies have explored diverse methodologies for emotion detection by integrating multiple modalities to improve the accuracy and applicability of emotion-aware systems. Techniques range from traditional facial expression analysis to cutting-edge deep learning algorithms.

- (1) **Convolutional Neural Networks (CNNs):** CNNs are extensively used for visual emotion recognition, processing facial expressions and image features with high accuracy. Arapakis et al. [12] employed CNNs for 3D facial analysis to enhance emotion detection in video recommendations, while Rumiantcev [146] used neural networks to predict emotions in valence-arousal space based on music features. Similarly, Krupa et al. [93] developed a two-level CNN-based system that integrates facial expression recognition and chatbot interactions, dynamically aligning music recommendations with users' emotions. Hossain et

Table 10. EMRSs categorization based on recommender system Algorithm

Categories	Algorithm	Publications
Traditional Models	SVM	[12, 25, 69, 89, 177]
	KNN	[20, 74]
	Random Forest	[20, 146]
	fuzzy inference	[196]
	CF	[89, 136]
	CBF	[136]
	Hybrid	[2, 132, 173, 177, 192]
Deep Learning Models	CNN	[1, 93, 132, 196]
	BERT	[203]
	Hybrid	[1, 9, 69, 146]

al. [69] utilized advanced CNN-based models, incorporating Mel-spectrograms for audio processing, 2D CNNs for speech recognition, and 3D CNNs with local binary patterns (LBP) and interlaced derivative patterns (IDP) for video emotion detection. Abdul et al. [1] applied DCNNs to extract latent emotional features from audio spectrograms and user metadata, correlating these features with mood and preferences to personalize music recommendations.

- (2) RNNs and LSTMs: RNNs and LSTMs capture temporal dependencies in speech and text-based emotion data, enabling systems such as the one proposed by Zhao et al. [203] to recognize conversational emotions effectively. Djordjević Čegar et al. [132] used neural networks to analyze physiological signals such as EEG, ECG, and EDA for estimating user emotions in the arousal-valence space. Additionally, Aivazoglou et al. [2] applied NLP techniques, including sentiment analysis, to textual and visual content for fine-grained emotion understanding.
- (3) SVM and Feature Optimization: SVMs are commonly utilized for emotion classification, particularly for audio features like pitch and MFCCs. Kim et al. [89] demonstrated the use of SVMs combined with collaborative filtering for multimedia recommendations. Genetic algorithms further enhance SVM performance by optimizing feature selection and improving classification accuracy.

10.2 Algorithms for Recommendation systems

Table 10 categorizes the algorithms used for EMRSs based on their respective methodologies.

10.3 Application Domains and Datasets

Emotion-aware multimodal recommender systems (EMRSs) utilize diverse datasets to capture and analyze emotional responses from sources such as images, audio, video, and physiological signals. These datasets enable the creation of systems that personalize recommendations by integrating multiple modalities. Below are key application domains, with Table 11 summarizing the datasets used in EMRSs.

- **Music:** EMRSs analyze inputs like facial expressions, speech patterns, and physiological signals to recommend music that matches the user's current mood. These systems dynamically adjust playlists, enhancing user satisfaction [9, 20, 146, 158].

- **Movies:** Multimodal data such as posters, soundtracks, and descriptions are utilized to align emotional content with user preferences. Techniques like fuzzy logic and the Jaccard similarity index address diverse emotional needs [196].
- **Social Media:** User-generated content, including posts and interactions, is leveraged to provide recommendations that resonate with users' emotional states. Sentiment analysis and interest profiling address challenges like content overload and cold-start problems [2].
- **POI:** EMRSs analyze user behaviors, such as check-ins and reviews, along with contextual data like location and time, to offer personalized POI suggestions. Hybrid models integrate preferences, spatial-temporal factors, and POI categories [192].
- **News:** Systems analyze user emotions based on past interactions to recommend content that positively influences their mood, using collaborative and content-based methods combined with real-time feedback [136].
- **Videos:** EMRSs analyze multimodal data like facial expressions, physiological signals, and metadata to recommend video clips aligned with users' emotional states, ensuring a tailored viewing experience [12, 74].
- **Mental Health:** Physiological signals such as ECG and GSR, combined with user interactions and conversations, help monitor emotional states. These systems recommend stress-reducing activities or content for mental health management based on real-time feedback [128].

Table 11. Most Commonly Used Datasets in EMRSs

Citation	Dataset	Explanation
[203]	IEMOCAP	Multimodal dataset with audio, video, and text for emotion recognition in dialogues.
[196]	TMDB and IMDb	Movie datasets with metadata for analyzing emotional responses to media.
[177]	IAPS	Image dataset designed to evoke emotional responses.
[51]	Author-gathered	Data collected from 30 participants watching five curated videos designed to elicit specific emotions

11 Addressing Research Questions

Having systematically reviewed emotion-aware recommender systems across six distinct modalities—from text-based sentiment analysis to multimodal fusion architectures—we now synthesize these findings by directly addressing the five research questions posed in Section 1. This synthesis integrates insights from modality-specific performance, ethical considerations, and deployment trade-offs to provide a unified understanding of the current state and future directions of EARS.

RQ1: Which emotional data modalities produce the largest improvements in recommendation accuracy? The integration of multimodal fusion pipelines consistently demonstrates reproducible improvements in various recommendation domains. Notably, in the movie domain, the incorporation of review-level emotional data significantly enhances recommendation precision. Saraswat et al. [155] showed that injecting emotions classified by a Bi-LSTM-CNN (Tweets Affective Classifier) resulted in a 173% increase in top-N precision compared to a sentiment-agnostic baseline.

Similar advancements are observed in music and video recommendation systems. Hossain et al. [69] reported that late fusion of audio Mel-spectrogram embeddings with video CNN features improved hit rate by 5–11 % over the best single-modal model.

While unimodal detectors, such as image or video-only systems, achieve high emotion-classification accuracy (e.g., ConvXGB on FER-2013 reaching approximately 73.28% accuracy [84]; Angkasa et al. [9], and facial CNNs achieving 85–90% accuracy), their translation into recommendation gains is often limited, typically less than 5%, and can be highly case-specific.

Furthermore, physiological cues, such as electroencephalography (EEG) or galvanic skin response (GSR), demonstrate the highest raw affect-recognition scores, ranging from 85–95% on datasets like DEAP/AMIGOS [77]. However, the practical deployment of such physiological sensors at scale remains rare, and consequently, the measured improvements in recommendation performance are largely anecdotal.

RQ2: What are the main ethical and privacy risks that arise from emotion-aware recommender systems? Three recurring risks emerge in the literature: (i) emotional manipulation, where systems exploit users' vulnerable moods for persuasive ends [174]; (ii) biometric-privacy breaches, because raw facial video or EEG/GSR traces are uniquely identifying and often captured passively [77]; and (iii) demographic bias, with emotion models trained on Western-centric images mis-classifying minority affect [174]. Reported mitigation strategies include federated or on-device learning to keep biosignals local (e.g., the emergency Health privacy-preserving pipeline [34]), differential-privacy noise injection for shared gradients, mandatory opt-in consent whenever wearables or cameras are used, and post-hoc fairness audits of affect classifiers to detect skew.

RQ3: How do efficiency and latency trade-offs vary between real-time and offline emotion extraction?

Real-time emotion extraction delivers adaptive moment-to-moment personalization but demands significant computational resources: multimodal pipelines with Vision Transformers, BERT, or streaming video analysis often exceed budgets for low-latency environments. Offline processing, in contrast, employs batch computation to apply more complex models and conduct extensive data analysis, typically achieving higher emotion detection accuracy; however, it does not reflect the current states of the users. The resulting trade-off has spurred research into lightweight model architectures, edge-computing deployments that shift inference closer to the user, and dynamic attention mechanisms that selectively activate only the most informative modalities, balancing immediacy against accuracy and scalability.

RQ4: What understudied themes offer the greatest opportunities for future research? Key gaps include cross-cultural variability [few studies evaluate how emotional expression differs across regions or languages] and longitudinal adaptation, where user emotional responses and preferences evolve over months or years. Interpretability remains an open challenge: explaining why a system inferred a particular emotional state or recommended content is vital for trust. Promising research fronts span advanced transformer-based emotion parsing for nuanced sentiment in dialogue, AR/VR integrations for immersive, affective experiences, and real-time federated multimodal models on resource-constrained devices. Interdisciplinary collaborations with psychology and ethics will be crucial to ensure fairness, privacy, and genuine user well-being as EARS matures.

RQ5: How do emotion-aware recommender systems perform across different data modalities in terms of emotion detection accuracy, recommendation effectiveness, and deployment feasibility? A comprehensive comparison of emotion-aware recommender systems across modalities remains elusive due to significant heterogeneity in datasets, evaluation metrics, and experimental setups. Each study often employs a distinct emotional dataset, ranging from physiological signals to text emotion corpora, making direct quantitative comparisons impractical. Likewise, evaluation methods vary widely: some research reports emotion recognition accuracy (e.g., F1, recall), others focus on recommendation relevance (e.g., CTR or NDCG), and few provide latency or resource usage metrics for deployment feasibility.

Furthermore, model architectures, including fusion techniques and feature extraction pipelines, differ substantially between modalities and cannot be easily normalized. This degree of methodological heterogeneity has been noted before in RS domains: inconsistent experimental practices hinder performance benchmarking and reproducibility efforts [59]. While quantitative synthesis remains limited, we provide a qualitative cross-modal analysis underscoring trade-offs:

- Text-based systems offer low-latency and high interpretability but suffer from limited emotional granularity;
- Video and physiological-based systems achieve richer emotional accuracy but at the cost of higher deployment complexity;
- Multimodal fusion generally enhances emotion detection and recommendation relevance, but demands significant computational investment and consistent evaluation frameworks.

This synthesis aligns with RQ3 in highlighting modality-specific strengths, limitations, and practical considerations in the design of EARS.

12 Discussions and Challenges

Emotion-aware recommender systems have significantly advanced personalized content delivery by leveraging affective computing, machine learning, and recommendation algorithms. These systems capitalize on their ability to interpret the emotional states of users, offering enriched user experiences in various domains such as movies, music, social media, and education. This review highlights both the substantial progress made and the areas that require further exploration, providing a comprehensive understanding of the current landscape.

The diversity in approaches used to detect and incorporate emotions into recommender systems reflects the field's maturity and adaptability. Text-based methods dominate due to the availability of data and advanced natural language processing techniques, while video, audio, and multimodal systems provide richer, context-aware insights by integrating diverse data modalities. Physiological data, although promising for its direct link to emotional states, remains underutilized due to challenges in data collection and interpretation. Emerging techniques such as transfer learning and multimodal fusion have begun to address these limitations, enhancing system capabilities and accuracy.

The field has also seen significant advances in real-time emotion tracking and adaptive systems. Wearable devices and improved sensor technologies are bridging the gap between static user data and dynamic emotional feedback, allowing for more personalized and responsive recommendations. For instance, recent architectures like MMER-LMF (Multi-Modal Emotion Recognition in Lightweight Modality Fusion) [85] strive to balance accuracy and computational efficiency in real-time scenarios. This lightweight fusion method achieves approximately 80% accuracy in emotion recognition, demonstrating superiority over single-modality approaches while ensuring adaptability and computational efficiency for practical deployment. Innovations in federated learning and decentralized architectures are addressing scalability and privacy concerns, offering ethical solutions for large-scale deployments without compromising user trust.

The applications of emotion-aware recommender systems extend beyond traditional entertainment domains to areas such as mental health support, personalized education, and adaptive user interfaces. These systems have the potential to improve user well-being and engagement by proactively addressing emotional needs. Although emotion-aware systems have shown promise in niche applications such as stress management apps (e.g., Calm, Headspace), their deployment in large-scale commercial recommender systems remains limited due to high computational costs, privacy concerns, and the complexity of real-time adaptation. Also, their broader social implications, such as the risks of reinforcing biases or unintentionally creating dependency on emotionally adaptive systems, require careful consideration.

Despite significant advancements in emotion-aware recommender systems, several unresolved issues persist across various modalities, including text, audio, image, video, physiological, and multimodal data.

In text-based systems, challenges such as data sparsity, cold start problems, and the difficulty of categorizing users with unique preferences remain pervasive. Ambiguity in opinion words and domain sensitivity complicate sentiment analysis, while scalability and the inherent subjectivity of opinions add further complexity. In addition, dynamic emotional states in long-form content and cultural variability in sentiment expression further hinder accurate emotion recognition, necessitating more adaptive and context-aware models.

Audio-based systems face hurdles in emotion recognition variability, dataset limitations, and the context dependency of music's emotional impact. Real-time adaptation in noisy environments and the subjectivity of user responses compound these challenges. Furthermore, distinguishing overlapping emotional categories like sadness and calmness, as well as accounting for speaker-specific variations in tone, pitch, and accent, remains a significant challenge for audio-based emotion detection.

Image-based systems struggle with data imbalance, achieving real-time performance, and ensuring accuracy across diverse cultural and individual contexts. Furthermore, decoding emotional expressions from visual content remains subjective and often ambiguous. The detection of subtle or micro-expressions, particularly in unconstrained settings with occlusions or poor lighting, further complicates the accuracy of emotion recognition systems.

Video-based systems face the high cost of data collection, the complexity of real-time emotion detection, and the lack of explicit emotional feedback. The interpretability and subjective nature of visual cues further challenge their adoption. Synchronizing emotional signals across facial expressions, gestures, and speech in dynamic video streams adds additional complexity, requiring careful temporal alignment and computationally efficient processing.

Physiologically-based systems rely on data such as heart rate, skin conductance, or brain signals. Challenges include non-intrusive yet accurate data collection, variability in individual physiological responses, and privacy concerns. Interpreting physiological data in a way that aligns with emotional states is an ongoing research area. Additionally, the non-stationary nature of physiological signals, coupled with the hardware limitations of wearable devices, restricts their effectiveness in real-world deployments.

For multimodal systems, key challenges include extracting meaningful representations from diverse modalities, effectively fusing heterogeneous features, and addressing data sparsity. The computational complexity of integrating large models, such as ViT, BERT, or LLM, makes these systems resource intensive and difficult to deploy in real time. The handling of noisy or incomplete multimodal data, aligning temporal and spatial information across modalities, and addressing biases in emotion recognition models remain critical issues. Furthermore, imbalanced modality contributions, where dominant inputs such as audio overshadow subtler signals such as physiological data, can degrade overall performance. Context dependency and the lack of interpretability in emotion-aware models further limit their adaptability and trustworthiness. Emotion-aware recommender systems raise significant ethical and privacy concerns due to the sensitive nature of emotional data. As Milano et al. systematically highlight, recommender systems can impose risks related to privacy, autonomy violations, bias, and lack of transparency [119]. Emotion-aware variants amplify these risks by harvesting deeply personal affective signals (e.g., facial expressions or physiological responses) that can be repurposed or analyzed without full user consent. Moreover, emotion recognition introduces the possibility of intentional or unintentional emotional manipulation, targeting vulnerable users at moments of sadness or stress to influence behavior beyond the relevance of recommendation. Recent proposals in affective computing ethics recommend explicit design safeguards such as data minimization, participant-informed consent, transparency, adaptive consent interfaces, and algorithmic accountability frameworks [22]. Ensuring cultural sensitivity, equity in emotional interpretation, and ongoing oversight will be fundamental to ethically aligned emotion-aware RS deployments.

13 Future Directions

Building upon the identified challenges, this section outlines four actionable research avenues that are critical to advance the field of Emotion-Aware Recommender Systems (EARS) in a responsible and impactful manner. These directions are identified to address the multifaceted concerns raised in our discussion, specifically focusing on user experience, ethical risks, system performance, and long-term validity.

13.1 Explainable Multi-Modal Fusion for EARS (addresses RQ1 & RQ2)

For EARS to foster trust and facilitate user control, their decision-making processes must be transparent. Future work should embed explainability directly into fusion architectures rather than retrofitting post hoc explanations, which often lack fidelity to the true model behavior. This inherent explainability is crucial for user adoption and ethical auditing, allowing stakeholders to understand why a particular recommendation is made and mitigating the risks of bias or manipulation [99].

Causal Fusion Layers. A critical advancement involves integrating causal-inference modules into each stage of the multi-modal fusion process. Traditional deep learning models often learn spurious correlations. By contrast, causal fusion layers would enable the model to identify and learn not just correlations but explicit causal links between specific multi-modal cues (e.g., a vocal tremor, a change in skin conductance, a subtle facial micro-expression) and inferred emotional states or user needs. This deeper understanding will allow EARS to provide richer, more robust explanations, moving beyond simple feature attribution to articulate the underlying "why" of its predictions. For instance, explaining "You seemed stressed because your heart rate increased sharply after that meeting and your voice showed signs of tension" provides a much clearer causal narrative than just "Your stress level is high" [87].

Modular Interpretability. We propose designing fusion networks with discrete, interpretable submodules, where each module is primarily responsible for processing and extracting insights from a single modality (e.g., one module for audio, another for visual, and yet another for physiological data). Crucially, these architectures would incorporate attention weights or other transparency mechanisms that can be surfaced as natural language explanations. This modularity not only aids interpretability by compartmentalizing complex processing but also facilitates debugging and auditing, as the contribution of each modality to the final recommendation can be clearly quantified and communicated. For example, an EARS could state that "The recommendation for calming music was primarily influenced by the high anxiety detected in your speech patterns, with secondary input from your elevated skin conductance."

Human-in-the-Loop Interfaces. To ensure explanations are truly useful and trusted, developing interactive human-in-the-loop dashboards is essential. These interfaces would allow users to actively query the EARS with questions such as "Why was this suggested?" or "What factors led to this emotion assessment?" More importantly, they would enable users to provide direct feedback on the explanations themselves – whether they found them clear, accurate, or helpful. This feedback mechanism would serve as a vital channel for continuously improving the explainability of EARS, closing the loop on transparency, and fostering a stronger sense of user agency and trust [100].

13.2 Federated & Differentially Private On-Device Learning (addresses RQ2 & RQ3)

The widespread collection of sensitive emotional and physiological data by EARS requires robust privacy-preserving mechanisms. Shifting the paradigm of model training from centralized servers to edge devices can simultaneously bolster privacy by keeping raw data local and significantly reduce latency, leading to more responsive and secure systems.

Optimized FedAvg. A key research direction involves adapting and optimizing the Federated Averaging (FedAvg) algorithm [26] for heterogeneous, continuous multi-sensor streams. Unlike discrete, static datasets, EARS operate on dynamic, high-dimensional data generated from diverse sensors (e.g., cameras, microphones, wearables) often with intermittent connectivity and unbalanced data distributions across user devices. Research must focus on improving communication efficiency, handling data drift, and developing aggregation strategies that are robust to non-IID (independent and identically distributed) data while maintaining model performance. This would enable EARS to learn from a vast distributed network of user experiences without compromising individual data privacy.

Adaptive DP Budgeting. Implementing context-aware differential privacy (DP) [54] for EARS is crucial. This involves developing mechanisms that dynamically tune the privacy parameter ϵ (epsilon) based on the sensitivity of the inferred emotional signal and specific user preferences. For instance, less noise might be intentionally added to aggregate updates related to a general mood state (e.g., "happy" or "calm"), while significantly more noise would be applied when dealing with highly sensitive or potentially stigmatizing emotional states (e.g., "anxious," "depressed," or "frustrated"), or when the user has explicitly set higher privacy preferences. This adaptive approach ensures a better balance between privacy protection and model utility, aligning privacy guarantees with the specific context of emotional data.

Secure Aggregation. To further enhance privacy beyond differential privacy, the use of Secure Multi-Party Computation (SMC) protocols [175] for the aggregation step in federated learning is vital. SMC techniques allow multiple parties (individual user devices) to collectively compute a function (e.g., averaging model updates) over their inputs while keeping those inputs private from each other and from the aggregation server itself. This means that even the central server would never see raw model updates in plain text, significantly mitigating the risk of data inference or reconstruction attacks, thus providing a stronger guarantee of data confidentiality at a crucial stage of federated training.

13.3 Embodied Continuous Self-Improvement (addresses RQ1)

To achieve truly personalized and adaptive user experiences, EARS must move beyond static, pre-trained models and embrace embodied continuous self-improvement. This paradigm shift enables systems to learn and adapt directly from ongoing, real-world interactions and experiences, reflecting the dynamic nature of human emotions and preferences over time.

Reinforcement Learning from Human Feedback. A promising direction is to take advantage of reinforcement learning (RL) frameworks where EARS receive continuous feedback from users within their contexts of embodied interaction. This feedback can be explicit (e.g., user corrections like "That recommendation wasn't helpful for my mood right now"), or, more powerfully, implicit, derived from physiological indicators of satisfaction or dissatisfaction (e.g., a decrease in stress markers after a positive recommendation, or an increase in cognitive load suggesting frustration). The RL agent would learn to refine its recommendation strategies based on maximizing user well-being, engagement, and satisfaction in real-time, effectively allowing the system to learn from its own "embodied" experiences and progressively align with individual user needs.

Meta-Learning for Adaptation. To facilitate rapid personalization, especially for new users or evolving scenarios, applying meta learning [58] (learning to learn) approaches to EARS is crucial. Meta-learning enables the system to quickly adapt to new users, contexts, or changing emotional landscapes with minimal additional data. Instead of training a model from scratch for each new user, a meta-learned EARS would possess a fundamental understanding that allows for efficient fine-tuning and high performance even with limited initial interaction data, making it highly flexible and personalized from the outset.

Lifelong Learning. For EARS to maintain their relevance and efficacy over extended periods, research must focus on lifelong learning algorithms [168] that effectively mitigate catastrophic forgetting. As systems continuously learn from new, diverse experiences, they must retain previously acquired knowledge (e.g., a user's long-term preferences, past emotional patterns) while seamlessly integrating new information. Robust lifelong learning ensures that EARS can accumulate knowledge and improve its understanding of individual users and emotional dynamics over months or years without losing past personalization or ethical safeguards, thereby offering sustained and reliable support.

13.4 Longitudinal & Cross-Cultural Evaluation Frameworks (addresses RQ4)

The validity, fairness, and sustained impact of EARS cannot be adequately assessed through short-term lab-based studies or evaluations confined to a single cultural context. Future research must develop and implement comprehensive longitudinal and cross-cultural evaluation frameworks to ensure the robustness and generalizability of EARS technologies. This addresses the critical need to understand how emotion recognition and recommendation effectiveness vary between diverse user groups and over extended periods.

Multi-Site Deployments. Conducting long-term, multi-site deployment studies in diverse geographic and cultural regions is paramount. Such studies would systematically measure how the accuracy of emotion recognition, the efficacy of recommendation, the engagement of users, and perceived utility vary between different cultures and demographics [94]. This goes beyond simple performance metrics to encompass user satisfaction, ethical perceptions, and the actual impact on user well-being over extended periods of time. For example, evaluating an EARS designed for stress management in both Western and Eastern cultural contexts would reveal critical differences in how stress is expressed, perceived, and responded to, leading to more culturally sensitive system designs.

Culture-Sensitive Annotation. The foundation of robust EARS lies in diverse and accurately annotated datasets. We must develop standardized protocols for emotion labeling that explicitly account for cultural expression differences. This involves collaborating with interdisciplinary experts (e.g., anthropologists, psychologists) to understand culturally specific nuances in emotional display rules, vocalic expressions, and physiological responses. Such efforts will ensure that training datasets reflect global diversity, preventing inherent biases in emotion recognition models that might otherwise arise from training predominantly on data from a single cultural group. This is vital for achieving equitable performance and avoiding misinterpretations between diverse user populations.

Impact Tracking Benchmarks. Current evaluation benchmarks often focus on short-term accuracy or immediate user feedback. Future research needs to create longitudinal impact tracking benchmarks that monitor user engagement, overall well-being, and ethical compliance over months or even years, rather than in one-off evaluations. These benchmarks should incorporate metrics that capture long-term behavioral changes, sustained positive emotional states, and the absence of negative ethical consequences (e.g., emotional manipulation, privacy breaches). This shift toward continuous, long-term monitoring will provide a more holistic and realistic assessment of EARS' societal impact, ensuring that the technology delivers sustained benefits while adhering to ethical principles.

13.5 Standardized Benchmarking and Evaluation Frameworks(addresses RQ5)

One key limitation observed across the surveyed literature is the lack of standardized benchmarks for evaluating emotion-aware recommender systems. Due to the diversity of modalities, datasets, and performance metrics (e.g., precision, recall, MAE, emotion classification accuracy), it remains difficult to draw direct comparisons between approaches. Future work should focus on curating shared benchmark datasets and unified evaluation protocols for different modalities (e.g., text, video, physiology) to enable reproducibility, fair comparison, and rigorous cross-system analysis. Such efforts could also facilitate leaderboard-based challenges to accelerate progress in the field.

14 Conclusion

Emotion-aware recommender systems mark a significant advancement in personalized content delivery by utilizing affective computing, machine learning, and multimodal data integration. By analyzing users' emotional states through text, audio, video, physiological signals, and multimodal, these systems enhance user experiences across various fields, including entertainment, education, healthcare, and e-commerce. This paper thoroughly

reviews the methodologies, datasets, challenges, and applications shaping the current landscape of emotion-aware recommender systems.

Despite significant progress, this field continues to face challenges, including data sparsity, real-time scalability, cross-modal fusion, and ethical considerations such as privacy, bias mitigation, and transparency. Emerging technologies, such as multimodal large language models, edge computing, and AR/VR integrations, offer promising avenues for addressing these challenges. Achieving their full potential requires fostering interdisciplinary collaboration, advancing methodologies for emotion detection and integration, and ensuring ethical guidelines are at the forefront. By doing so, emotion-aware recommender systems can continue to evolve responsibly, delivering empathetic, adaptive, and impactful solutions across industries while aligning with principles of fairness, transparency, and user trust.

15 Acknowledgments

This work is partially sponsored by Natural Science and Engineering Research Council of Canada (Grant 2020-04760).

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Received 23 December 2024; revised 28 January 2026; accepted 14 April 2026